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GRADUATION PROJECT

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1. Abstract

1.1 Efficiency Enhancement in Fiber Laser Cutting Process: A Graduation on Compressed Air Usage and Parameter Optimization for 2mm Mild Steel

Increasing competitiveness in the metal processing industry is possible by reducing production costs and increasing processing speed. This graduation project aims to increase efficiency by examining the cutting processes of specifically 2mm thick mild steel materials in a production line that cuts an average of 50 sheets per day.

In the current state, it has been observed that cuts made with Nitrogen (N₂) gas on a Bodor fiber laser machine are costly and suffer from dross formation problems due to incorrect parameters. Within the scope of the project, the technical and economic feasibility of using Compressed Air instead of expensive Nitrogen gas has been investigated through literature reviews and machine parameter analyses.

In light of the theoretical analyses and parameter optimization studies (negative focusing and high-pressure strategy), it is projected that the production speed can be increased by 25% if the compressed air cutting method is adopted. Furthermore, it is concluded that applying the proposed new parameter set will eliminate the current dross problem, improve surface quality, and significantly reduce the unit part cost.

This study presents the necessary technical infrastructure and parameter recipe for cost-effective and high-speed cutting in the mentioned production line.

2.Information About Dalgakıran Makina San. ve Tic. A.Ş.:

2.1 History, Overview, and Global Position

Dalgakıran Compressor, Turkey's largest industrial compressor manufacturer and exporter, started its operations in 1965 by Ömer Dalgakıran in a humble 25 m² workshop in Istanbul. Growing rapidly with investments made since its foundation, the company today leads its sector with over 900 employees and modern production facilities covering a total indoor area of over 70,000 m².

Carrying Turkish engineering to the global market with over 60 years of experience and a dynamic staff, Dalgakıran is ranked among Turkey's top 500 exporters. The company provides added value to industrial development worldwide with exports to over 130 countries, a network of over 150 dealers, and widespread service points. One of the most important factors reinforcing its leading position in the sector is possessing Turkey's first R&D center in this field and combining technology with innovation.



Figure 1. Dalgakıran Makina San. ve Tic. A.Ş. Factory

2.2 Vision and Mission

The company's core vision is to maximize customer satisfaction as the most reliable, innovative, and high-quality compressor manufacturer in Turkey and the region. In light of this vision, its mission is to produce high-quality, durable, and energy-efficient compressor systems, to create a difference in the global market through continuous improvement and technological innovation, and to develop sustainable, eco-friendly solutions.

2.3 Production Facilities and Plant Layout

Dalgakıran production facilities are designed to optimize the process from raw material entry to final product testing according to "Lean Manufacturing" principles. The factory layout consists of the following main sections:

- **Manufacturing Units:** Machining, Welding Shop, Casing Manufacturing, and Paint Shop.
- **Assembly Lines:** Specialized "Ant Line," "Sparrow Line," "Eagle Line," and "Owl Line" organized to ensure efficient workflow.
- **Support Units:** Electrical Workshop, Sponge Workshop, Storage, and Test Room.

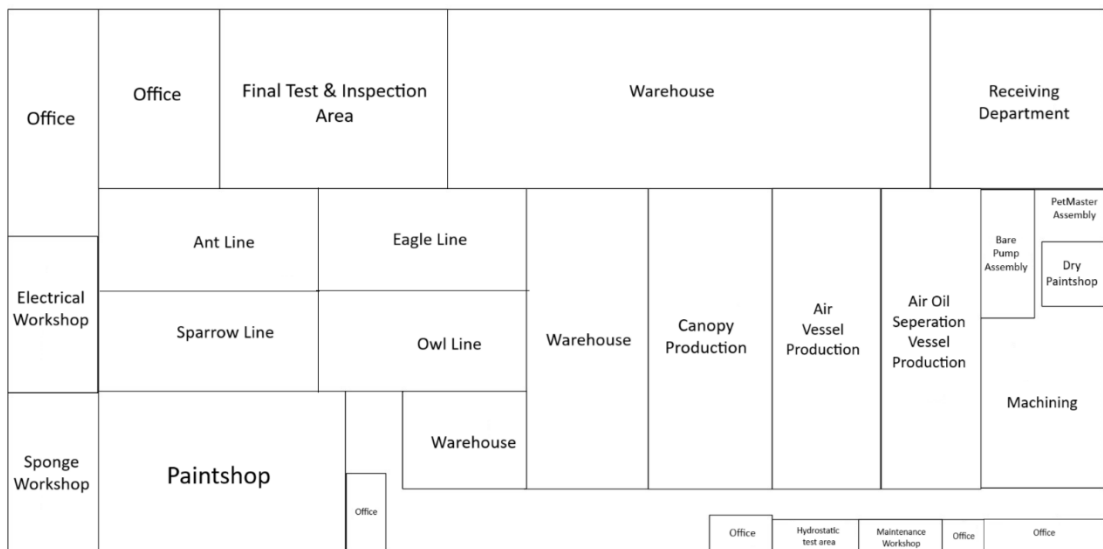


Figure 2. Layout Plan of the Facility

2.4 Product Range and Technical Competencies

Dalgakıran offers a wide and high-tech product range tailored to different industrial needs:

2.4.1 Screw Compressors (e.g., Impetus VSD): Industrial units that provide high energy efficiency, feature inverters, operate quietly, and consist of screw blocks and cooling systems.

2.4.2 Reciprocating (Piston) Compressors: Systems with cylinder-piston-crank mechanisms designed specifically for businesses with fluctuating air demand.

2.4.3 Oil-Free Compressors (e.g., PetMaster Series): High-performance products providing 100% oil-free air, manufactured for sensitive sectors such as food and health.

2.5 Organizational Structure, Production Culture, and Technology

Possessing a large organizational structure, management at Dalgakıran is divided into directorates such as Production, R&D, Supply Chain, Finance, Human Resources, and Operational Excellence under the Board of Directors and the Management and System Presidency. Production activities are carried out under the Production Directorate, led by the "Production Manager" and "Lean Production Manager."

A culture of work discipline and continuous improvement (Kaizen) prevails in the factory. Processes are inspected on-site through "GEMBA" walks held every morning with the participation of production managers. Regarding technological infrastructure; SolidWorks is used for product designs, while material tracking and technical drawing controls are managed digitally via the SAP (ERP) system. Additionally, processes are continuously optimized using advanced manufacturing methods such as semi-automatic welding machines and laser cutting technologies.

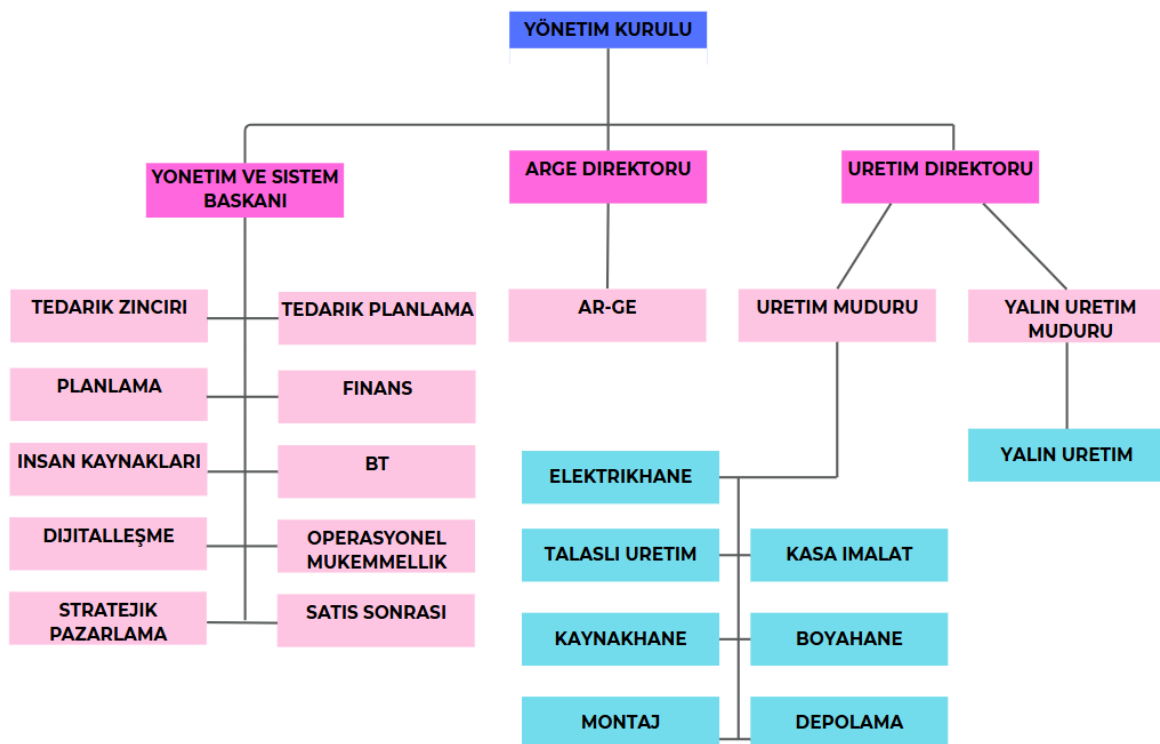
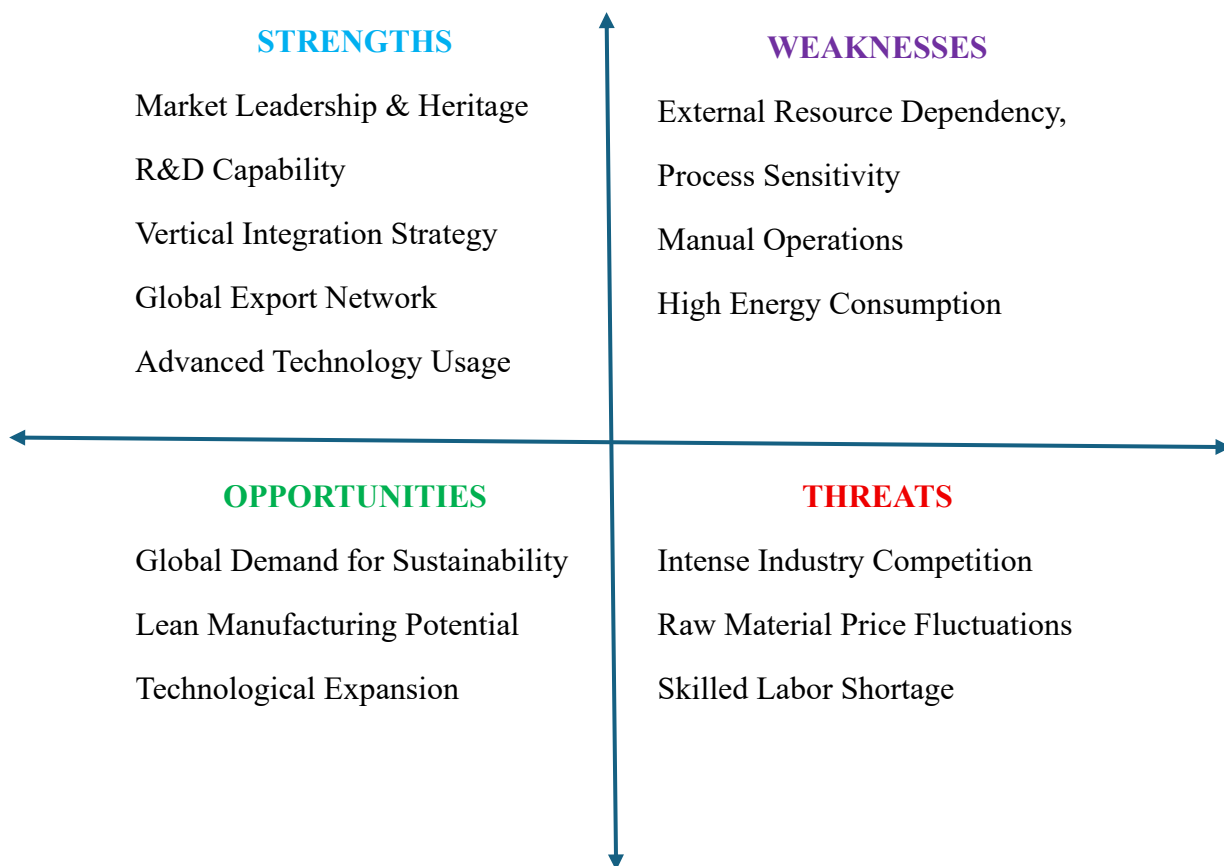


Figure 3. Organizational Structure

2.6. Strategic Analysis: SWOT Assessment of Dalgakıran Kompresör

This section presents a strategic evaluation of Dalgakıran Kompresör, analyzing its internal strengths and weaknesses alongside external opportunities and threats. As Turkey's largest industrial compressor manufacturer, the company's strong R&D infrastructure and vertical integration model are highlighted against the operational challenges inherent in large-scale manufacturing.



STRENGTHS

1. **Market Leadership & Heritage:** Founded in 1965, the company is Turkey's largest industrial compressor manufacturer and exporter, operating in a 70,000 m² facility with over 900 employees.
2. **R&D Capability:** Possessing Turkey's first R&D center in this field allows the company to combine technology with innovation and develop high-efficiency products.
3. **Vertical Integration Strategy:** The company adopts a vertical integration model by manufacturing compressor canopies (casings) in-house rather than outsourcing, ensuring total quality control.
4. **Global Export Network:** With exports to over 130 countries and a network of 150+ dealers, the company has a strong and diversified income structure against local market fluctuations.
5. **Advanced Technology Usage:** The production line utilizes advanced manufacturing technologies such as 12kW Fiber Lasers, semi-automatic welding machines, and SAP (ERP) systems for digital tracking.

WEAKNESSES

1. **External Resource Dependency:** While canopy manufacturing is done in-house, the process relies heavily on external suppliers for critical consumables (such as cutting gases), creating a "Variable Cost" burden.
2. **Process Sensitivity:** Inefficiencies or downtimes at the initial stages of production (like laser cutting) directly impact all subsequent bottlenecks such as bending, welding, and painting.
3. **Manual Operations:** despite high automation, certain intermediate processes still require manual labor, which can lead to variability in unit times and "Non-Value Added" activities.
4. **High Energy Consumption:** Operating heavy machinery parks (Laser, Press Brakes, Welding Robots) results in significant energy consumption, necessitating strict cost management.

OPPORTUNITIES

1. **Global Demand for Sustainability:** The increasing global demand for energy-efficient and eco-friendly industrial solutions aligns with the company's mission to develop sustainable products.
2. **Lean Manufacturing Potential:** The established "Lean Manufacturing" culture and regular GEMBA walks provide a continuous opportunity to identify and eliminate wastes (Muda) in the production line.
3. **Technological Expansion:** The R&D center offers the opportunity to develop new product lines (e.g., Oil-free compressors like PetMaster) for sensitive sectors like food and health.

THREATS

1. **Intense Industry Competition:** The metal processing and machinery industry faces increasing pressure to reduce production costs and increase processing speeds to maintain competitiveness.
2. **Raw Material Price Fluctuations:** As a manufacturer relying on steel procurement, global volatility in raw material prices poses a risk to profit margins.
3. **Skilled Labor Shortage:** The operation of high-tech machinery requires highly qualified personnel, and finding or retaining such talent is a general industrial challenge.

3. Selection and Justification of the Project Topic

In order to determine the most suitable project topic that will create added value in Dalgakıran Compressor production processes, bottlenecks and improvement areas throughout the facility were examined. Three potential project alternatives identified were subjected to a detailed technical and economic evaluation, considering critical performance indicators such as technical feasibility, investment cost, and return on investment (ROI).

3.1. Alternative Project Proposal 1: Robotic Welding Cell Integration for Air Receivers

1. Current Situation and Problem Definition

In Dalgakıran Compressor production facilities, "Air Receiver Tanks," which are critical components of screw and piston compressors, are manufactured in the internal "Welding Shop." In the current process, the joining operations of tank bodies and dished ends are performed manually by certified welding operators using the Gas Metal Arc Welding (GMAW/MIG-MAG) method.

During field observations, the following problems were identified:

- **Quality Variability:** Depending on operator fatigue and competence, variations occur in weld seam quality (penetration and visual appearance), leading to rework.
- **Health and Safety (HSE) Risks:** Exposure to welding fumes and intense UV radiation poses long-term health risks for operators.
- **Cycle Time Instability:** During the circumferential welding of large-volume tanks (500L - 1000L), the production speed decreases as the operator needs to frequently change positions and rest.

2. Proposed Solution: 6-Axis Robotic Welding Cell

It is proposed to transform the existing manual station into a fully automated Robotic Welding Cell.

- **System Components:** Integration of 1 x 6-axis industrial robot arm (e.g., Fanuc/Kuka), 1 x synchronized rotary positioner, and a high-amperage digital welding power source.
- **Operation Principle:** The operator will only load the part onto the fixture. The robot, synchronized with the positioner, will perform the welding operation continuously and with standardized quality.

3. Expected Benefits

- **Standardization:** 100% standard, defect-free weld seams capable of passing Non-Destructive Testing (NDT/X-Ray).
- **Capacity Increase:** A projected 40% increase in daily production capacity by stabilizing welding speed.
- **HSE Improvement:** Minimizing occupational risks by removing the operator from the immediate source of fumes and radiation.

4. Feasibility and Decision (REJECTED)

As a result of the cost-benefit analysis, this project was eliminated for the following reasons:

- **High Investment Cost (CAPEX):** The initial setup and hardware cost is estimated to be in the range of €60,000 - €80,000.
- **Long Return on Investment (ROI):** The calculated ROI period exceeds 2.5 years, which is considered too long for a short-term efficiency project.
- **Implementation Complexity:** The requirement to stop the production line during installation and the need for specialized personnel for robot programming were deemed restrictive.

3.2. Alternative Project Proposal 2: Lean Layout Optimization for Assembly Line

1. Current Situation and Problem Definition

In the assembly line designated as the "Sparrow Line," where small and medium-sized screw compressors are assembled, inefficiencies in material flow and station layout were observed. Assembly operators travel excessive walking distances throughout the shift to collect parts and merge sub-assembly groups.

The main wastes (Muda) identified in the current layout are:

- **Transportation Waste:** Storage of side panels and electrical cabinets at points far from the assembly bench.
- **Motion Waste:** Operators constantly bending down or taking unnecessary steps between stations during assembly.
- **Line Imbalance:** Bottlenecks occurring at certain stations while others remain idle.

2. Proposed Solution: U-Type Cell and Spaghetti Diagram Analysis

It is proposed to redesign the assembly line using Lean Manufacturing techniques.

- **Methodology:** Mapping current operator movements using a Spaghetti Diagram to identify and eliminate "Non-Value Added" movements.
- **Layout Redesign:** Transitioning from a straight line (I-Type) layout to a U-Type Assembly Cell model, where operators can assist each other, and material replenishment can be done from the outside without interrupting the flow.
- **5S Implementation:** Positioning hand tools and fasteners within the operator's "Golden Zone" (immediate reach).

3. Expected Benefits

- **Efficiency:** A projected 30% reduction in operator walking distances.
- **Space Saving:** Approximately 15% floor space gain in the factory production area.
- **Ergonomics:** Reducing operator fatigue and lowering the risk of musculoskeletal disorders.

4. Feasibility and Decision (REJECTED)

Although this project requires "Zero Investment," it was eliminated for the following reasons:

- **Quantification Difficulty:** The gain to be achieved is primarily "labor time" savings (Soft Savings). It is difficult to measure this as direct cash flow (Hard Savings) entering the company budget compared to material cost reductions.
- **Limited Impact Scope:** The improvement covers only the assembly line and does not affect the upstream processes (cutting, bending, painting) that dictate the overall factory pace.
- **Time Constraint:** Completing the layout changes and ensuring operator adaptation within the 12-week project period poses a risk to finalizing the results.

3.3. Selected Project Proposal 3: Parameter Optimization and Use of Compressed Air in Laser Cutting

1. Current Situation and Problem Definition

At the "Laser Cutting" station, which is the starting point of the Dalgakıran Compressor production process, Liquid Nitrogen (N_2) is used for cutting thin sheets (1-3mm Mild Steel) utilized in casing manufacturing. Examinations on the facility's busiest machine, the Bodor 12kW Fiber Laser, revealed two critical problems:

1. **High Cost:** Liquid nitrogen procurement constitutes a massive "Variable Cost" item, exceeding 1.1 Million TL annually for a single machine.
2. **Quality Problem (Dross):** With current nitrogen parameters, due to incorrect focus selection (positive focus) and insufficient gas pressure, heavy dross forms on the bottom surface of the sheets, necessitating additional manual labor (grinding/deburring).

2. Proposed Solution: Cutting with High-Pressure Compressed Air

It is proposed to use 16 Bar Compressed Air as the assist gas, generated using the company's own compressor technology, instead of expensive nitrogen gas.

- **Technical Approach:** Utilizing a hybrid method that leverages the exothermic energy of the 21% Oxygen in atmospheric air along with the kinetic cooling effect of the 78% Nitrogen.
- **Parameter Optimization:** Shifting the focus point into the material (Negative Focus, -1.5mm) and switching the nozzle configuration to a "Single Nozzle" design to enhance gas dynamics.

3. Expected Benefits

- **Cost Savings:** An over 80% reduction in assist gas costs (dropping from 570 TL/hour to 98 TL/hour).
- **Speed Increase:** A 25% increase in cutting speed (from 20 m/min to 25 m/min) due to exothermic support.
- **Process Improvement:** Eliminating the dross cleaning operation, thereby aligning with "Lean Manufacturing" goals by removing non-value-added processes.

4. Feasibility and Decision (ACCEPTED)

As a result of the comparative analysis, this project was selected as the primary project for the following reasons:

- **Rapid ROI:** The return on investment period is calculated to be 3.6 months, which is significantly below industrial averages.
- **High Impact:** The savings directly reduce cash expenses (OPEX), and the results are immediately reflected in financial reports.
- **Strategic Fit:** It is a prestigious example of corporate identity and engineering capability for a compressor manufacturer to utilize "Compressed Air" in its own production to achieve efficiency.

3.3. Project Selection Decision Matrix (Pugh Matrix)

Evaluation Criteria	Weight	1. Laser Cutting Air Conversion (Selected)	2. Robotic Welding Integration	3. Assembly Line Optimization
Investment Cost (Lower is better)	30%	9/10 (Only compressor investment)	2/10 (High robot cost)	10/10 (No investment required)
Return on Investment (ROI)	30%	10/10 (3.6 Months - Very Fast)	3/10 (2 Years - Slow)	6/10 (Uncertain/Long)
Ease of Implementation	20%	8/10 (Can be done without stopping machine)	4/10 (Line stoppage required)	5/10 (Physical relocation is hard)
Cost Saving Impact	20%	10/10 (Direct cash savings)	7/10 (Labor savings)	6/10 (Efficiency increase)
TOTAL SCORE	100%	9.3 (WINNER)	3.8	7.2

As seen in table, although the 'Assembly Line Optimization' project had the advantage of zero investment, it scored lower due to the difficulty in quantifying its financial impact. The 'Robotic Welding' project was eliminated due to its high initial cost and long ROI period. Consequently, the 'Laser Cutting with Compressed Air' project achieved the highest total score of 9.3, primarily driven by its rapid return on investment and significant, measurable cost reduction potential.

3.4. Project Proposal 3 Selection and Justification:

Critical Point for Cost Analysis: In the casing manufacturing process, the "Laser Cutting Operation" is the most distinct and measurable input that directly affects the unit cost. While other stages are predominantly based on labor and machine depreciation, the laser cutting process is the area most open to Cost Reduction due to the use of externally supplied Assist Gas (Nitrogen/Oxygen).

External Resource Dependency: The procurement of gases used in cutting machines from external suppliers turns this item directly into a "Variable Cost". Reducing this consumption or converting it to a more economical source (Compressed Air) translates to direct cash savings for the enterprise.

Machine Park and Working Schedule: There are 3 laser cutting machines actively operating within the facility. These machines are operated at an intense pace in a 2-shift (Night/Day) arrangement to meet production demand. This intensity indicates that even the smallest improvement in unit time (seconds) or unit cost will transform into a significant gain on an annual basis due to the multiplier effect.

Observation Methodology: To ensure the reliability of project data, the Day Shift was selected as the study period. Among the 3 machines, the machine with the highest workload and part variety (Bodor 12kW) was designated as the pilot area, and the analyses were based on the production data from this machine.

In conclusion; it was observed that inefficiencies in the cutting stage, which is the first operation following raw material entry, affect all subsequent steps (bending, welding, painting). Therefore, it was decided that this is the most strategic point for the "Cost-Efficiency" study.

4. Scope and Limitations of the Project

This graduation project was conducted to enhance the efficiency of sheet metal cutting operations at the Dalgakıran Compressor production facility. The technical framework and boundaries of the study are outlined below:

4.1 Scope

4.1.1 Machinery: The study covers tests and observations performed on the 12kW Bodor MFSC12000 Fiber Laser Cutting Machine, which has the highest workload in the active production line.

4.1.2 Material Type: The project focuses on Low Carbon Steel (Mild Steel / S235JR) sheets used in compressor canopy manufacturing.

4.1.3 Material Thickness: The investigation is limited to 1 mm, 1.5 mm, and 2 mm sheet thicknesses, which constitute the highest circulation in the production line.

4.1.4 Process Variables: A comparative analysis was conducted between the currently used Oxygen (O₂) / Nitrogen (N₂) gases and the proposed Compressed Air cutting parameters (Speed, Pressure, Focus Point, Nozzle Type).

4.1.5 Data Collection: Observations are based on production data collected during the Day Shift, when machine usage and operator performance are at their peak.,

4.2 Limitations

4.2.1 Material Variety: Other metal types such as Stainless Steel, Aluminum, and Brass are excluded from the scope of this study.

4.2.2 Thickness Limit: The efficiency and edge quality of air cutting for thicknesses exceeding 2 mm are not examined in this project; the focus is strictly on thin sheet applications.

4.2.3 Test Method: Cutting quality evaluations are limited to visual inspection (dross check), surface roughness observation, and cutting speed (m/min) data. Microscopic metallurgical examination or mechanical tensile/rupture tests were not performed on the samples.

4.2.4 Machine Park: Although there are two other laser cutting machines in the facility, they were not included in this analysis process to ensure data consistency; all data was standardized based on the 12kW Bodor machine.

5. Current Situation Analysis

5.1 Field Study:

The Bodor MFSC12000 (12kW) fiber laser cutting station, which is actively operating and possesses the highest workload within the Dalgakıran Compressor production facilities, was selected as the application site for the project. The following steps were followed during the data collection process:

5.1.1 Observation Period: The Day Shift working hours (07:30 - 17:30), where machine and operator performance could be observed most clearly, were used as the basis.

5.1.2 Data Collection: Dross formation problems occurring in cuts made with existing Nitrogen (N₂) and Oxygen (O₂) parameters, cutting speeds, and the preparation times spent by operators at the machine were recorded using the "Time Study" method².

5.1.3 Process Analysis: Cutting-related bottlenecks and cost elements were determined using workflow charts in the production line and findings from GEMBA walks³.

5.2 Theoretical Parameter Optimization

In light of the obtained field data and literature knowledge, a new parameter set was designed to reduce costs and increase speed. The targeted optimization strategy for the Bodor machine's control unit includes the following:

5.2.1 Focus Position: Lowering the focus below the material surface (negative focus) to facilitate melt evacuation in air cutting.

5.2.2 Gas Pressure: Increasing the pressure to levels of 14-16 Bar to rapidly remove molten metal from the cutting zone.

5.2.3 Nozzle Selection: Determining the Single Nozzle configuration instead of Double Nozzle to improve airflow dynamics.

5.2.4 Cutting Speed: Increasing the cutting speed by 20-30% by utilizing the exothermic contribution of oxygen in the air.

With this method, the aim was to create the most efficient production recipe based on scientific data while minimizing trial-and-error costs.

6. Methodology of the Study

This graduation project was designed and conducted following a systematic, three-stage methodology comprising a comprehensive literature review, a current situation analysis based on empirical field observations, and a theoretical parameter optimization phase. This structured approach ensures that the project is not merely based on trial-and-error but is grounded in scientific principles and validated by real-world industrial data.

- **Phase 1: Literature & Theoretical Framework:** The study began by establishing a strong theoretical foundation regarding fiber laser physics, gas dynamics, and material science to understand the interaction between the 12kW laser beam and 2mm mild steel.
- **Phase 2: Field Analysis (Data Collection):** Real-time production data was collected from the Dalgakıran manufacturing line, focusing on cycle times, gas consumption rates, and quality metrics of the existing Nitrogen/Oxygen processes.
- **Phase 3: Optimization & Validation:** Finally, a new parameter set was developed using the gathered data, focusing on maximizing the thermodynamic advantages of compressed air while minimizing common defects like dross and surface roughness.

6.1 Literature Review

In the first phase of the study, the effects of assist gases on cutting quality, processing speed, and operational cost in fiber laser cutting technology were examined in detail. Specifically, the technical and economic differences between the use of Traditional Nitrogen (N₂) and Compressed Air were investigated for the cutting of 1-2 mm thin-walled carbon steel (mild steel) materials using 12kW high-power laser sources.

By reviewing academic sources, peer-reviewed journals, and machine manufacturer technical bulletins, a theoretical framework was established. This framework focuses on three core areas:

6.1.1 Thermodynamic Effects: How the 21% oxygen content in compressed air provides an additional exothermic energy boost compared to inert nitrogen.

6.1.2 Oxidation Process: The formation of the oxide layer on the cut edge and its impact on subsequent processes like painting or welding.

6.1.3 Cost Efficiency: The economic advantage of generating assist gas on-site via a compressor versus the recurring procurement costs of bulk liquid nitrogen.

6.2 Related Studies in the Literature

The effects of varying assist gases (Nitrogen, Oxygen, and Air) on cutting quality and cost in the laser cutting process have been extensively investigated in academia across various material types. Two significant studies that are directly related to our specific project scope and material parameters are summarized below:

6.2.1. Optimization of Cutting Parameters (Patel and Bhavsar, 2021)

In the study published by Patel and Bhavsar in the journal *Materials Today: Proceedings*, parameters affecting cutting quality in the fiber laser cutting process were analyzed using Response Surface Methodology (RSM). Rather than providing a single static speed value, the researchers focused on the statistical impact of the complex relationship between Laser Power, Gas Pressure, and Cutting Speed on dross formation. They highlighted that the interaction between these variables is non-linear and requires precise balancing.

- **Relevance to Our Project:** This study is pivotal for our research as it demonstrated that gas pressure is the most critical variable in evacuating molten material from the cutting kerf (channel). The study reported that optimum gas pressure combined with precise focus settings significantly minimized dross height. This finding supports the theoretical validity of the "High Pressure (16 Bar) and Negative Focus" strategy adopted in our project. By lowering the focus point, we widen the kerf slightly to allow the high-pressure air to enter and eject the melt before it solidifies, a concept validated by Patel and Bhavsar's findings.

6.2.2. Analysis of Air Cutting on Different Materials (IYTE Thesis)(2023)

In a thesis study conducted at the Izmir Institute of Technology (IYTE), fiber laser cutting parameters were investigated on DIN 1.2379 (AISI D2) Cold Work Tool Steel. This material is widely used in industrial die making and is known for its high hardness and difficulty to machine compared to standard mild steel. The study provided a comparative analysis of surface roughness and cutting costs.

- **Findings:** The thesis observed that even with the 1.2379 material, which possesses a hard, high-carbon structure and high thermal resistance, the use of Compressed Air as an assist gas significantly reduced operational costs compared to Oxygen and Nitrogen gases without causing critical quality failures.
- **Relevance to Our Project:** This study serves as a "stress test" validation for our proposal. Since the Mild Steel used in our study has a much lower carbon content and hardness compared to the 1.2379 tool steel used in the thesis, oxidation and surface roughness behaviors during air cutting are predicted to be at a much more manageable level. This literature reference reinforces the feasibility of this method for softer sheets (2mm Mild Steel) by demonstrating that if air cutting is effective and economical even for difficult-to-machine steel groups, it is undoubtedly a viable solution for our standard production materials.

7. Theoretical Background and Literature Review

7.1. Fiber Laser Cutting Technology

Fiber lasers are solid-state laser types that have revolutionized industrial metal processing in the last twenty years. Unlike CO₂ lasers, beam generation in fiber lasers occurs within an active optical fiber doped with rare-earth elements (e.g., Ytterbium - Yb). This technology is preferred due to its high energy efficiency (30%-40% wall-plug efficiency), excellent beam quality (BPP - Beam Parameter Product), and low maintenance costs.

7.1.1. Working Principle

The working principle of a fiber laser is based on the "Stimulated Emission" mechanism. The system consists of the following stages:

- **Pumping:** High-power laser diodes stimulate the Ytterbium ions inside the active fiber, raising them to a higher energy level.
- **Resonance and Amplification:** The stimulated atoms emit photons. These photons oscillate between the Bragg gratings (mirrors) at the ends of the fiber, gaining intensity.
- **Beam Delivery:** The generated laser beam, with a wavelength of approximately **1.064 um**, is flexibly transmitted to the cutting head via a passive delivery fiber.
- **Focusing:** The collimator and focusing lenses in the cutting head focus the beam onto the material surface to a micron-level spot (e.g., 50-100 um). At this point, the energy density reaches levels of MW/cm² instantaneously melting or vaporizing the material.,

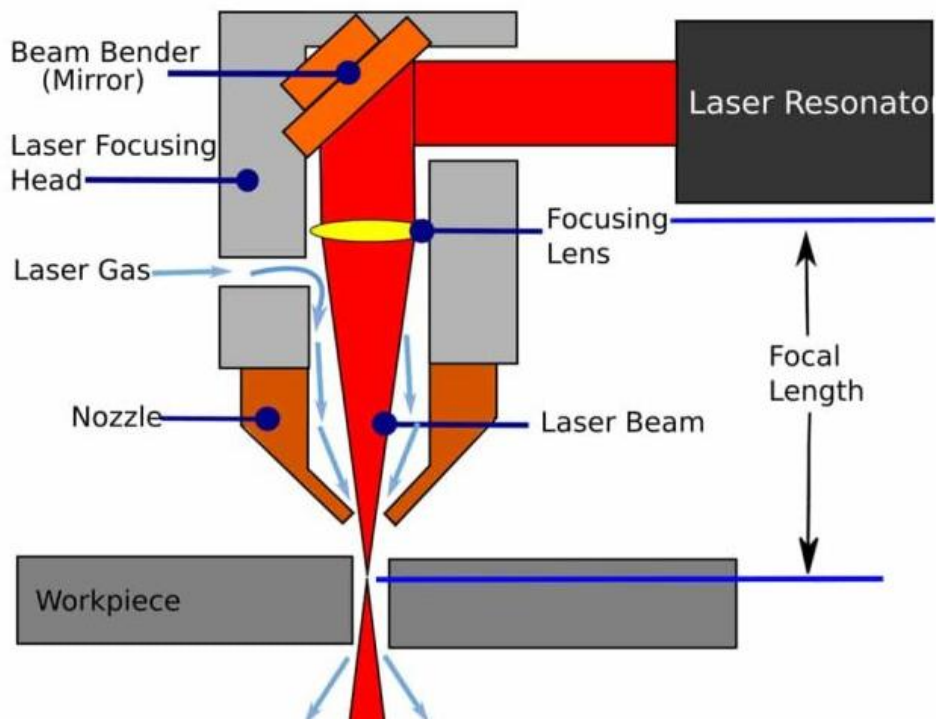


Figure 4. Fiber Laser Working Diagram

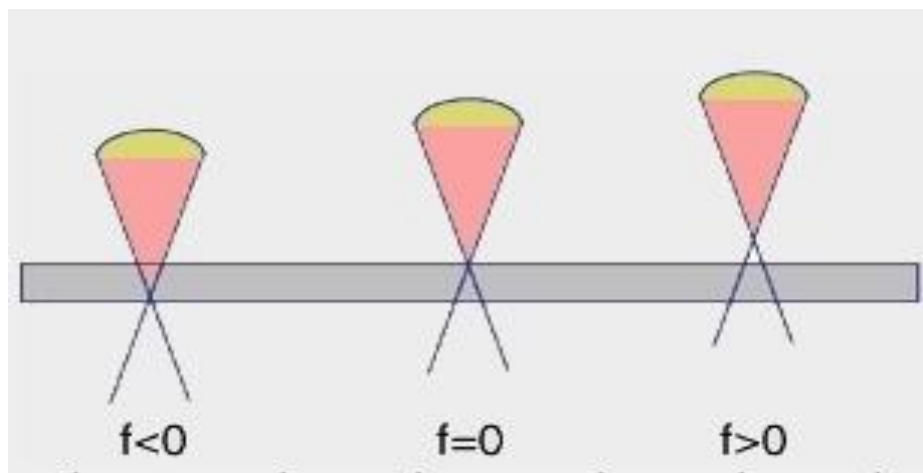


Figure 5. Focus Position Visualization

7.1.2. Effect of Process Parameters

Cutting quality and speed depend on the balance of fundamental process parameters:

- **Laser Power:** Determines the total energy transferred to the material. High powers, such as the **12kW** used in this project, allow energy to reach the lower layers of the material rapidly. As power increases, cutting speed can be increased; however, excessive power (at constant speed) causes overheating of the material and the formation of a wide kerf (cutting gap).
- **Cutting Speed:** It is inversely proportional to the Heat Input ($H = P / v$).
 - *Low Speed:* Causes plasma formation and overburning of the material.
 - *High Speed:* The beam cannot fully melt the material or the gas cannot eject the melt, leading to "cut loss" or drag lines.
- **Frequency and Duty Cycle:** Refers to whether the laser operates in Continuous Wave (CW) or Pulse mode. In thin sheet cutting, CW mode is generally preferred, while frequency modulation is used to control heat accumulation at corners and complex geometries.

7.2. Thermodynamic Role of Assist Gases in Laser Cutting

The laser beam melts the material, but the element that completes the cutting process (by removing the melt) is the assist gas. The type of gas radically changes the thermodynamic process.

7.2.1. Cutting with Nitrogen (N₂) (Inert Cutting)

Nitrogen acts as an inert (noble) gas, isolating the cutting zone from atmospheric oxygen.

- **Mechanism:** Applies only mechanical thrust force. There is no chemical reaction (it has an **Endothermic** cooling effect).
- **Advantage:** The cut surface is bright; no oxide layer forms. Sandblasting is not required before painting.
- **Disadvantage:** The laser must provide all the heat required for cutting, so it is slow on thick materials. Additionally, the unit cost per m³ is high.

7.2.2. Cutting with Oxygen (O₂) (Reactive Cutting)

Oxygen enters into an exothermic reaction with iron (Fe).

- **Mechanism:** $\text{Fe} + \text{O}_2 \rightarrow \text{FeO} + \text{Heat}$. This reaction adds 40%-60% extra chemical energy to the system in addition to the laser energy.
- **Advantage:** Thick materials can be cut with low laser power. Gas pressure is low (0.5 - 6 Bar).
- **Disadvantage:** A black, brittle oxide layer forms on the cut surface (preventing paint adhesion). Cutting speed is limited by the speed of the combustion reaction; it is slow on thin sheets.

7.2.3. Use of Compressed Air and Thermodynamic Effects

Compressed air (~78% Nitrogen, ~21% Oxygen) is a hybrid version of Nitrogen and Oxygen cutting.

- **Thermodynamic Effect:** The 21% Oxygen content initiates a mild exothermic reaction in the melt pool. This reaction allows for higher cutting speeds compared to pure Nitrogen cutting.
- **Kinetic Effect:** The 78% Nitrogen content and high pressure (14-16 Bar) cool the formed plasma and eject the melt from the kerf channel with strong momentum (shear force).
- **Result:** In 1-2 mm sheets, Air cutting is approximately **2 times faster** than Oxygen cutting and **70% more economical** than Nitrogen cutting.

7.3. Cutting Defects in Thin Sheets

Incorrect selection of laser cutting parameters (especially focus and gas pressure) leads to characteristic defects in thin materials.

7.3.1. Dross Formation

Dross is the residual metal that solidifies at the bottom edge of the cutting kerf and adheres to the part. The formation mechanism is the competition between the viscosity of the molten metal and the sweeping force of the gas.

- **Low Speed Dross:** The material overheats, the melt becomes very fluid, and bead-like soft dross forms.
- **High Speed Dross:** The laser head advances before the gas can eject the melt. The melt "lags behind," forming sharp, hard dross (drag dross).
- **Pressure-Induced Dross:** As observed within the scope of the project, in cases where air pressure is insufficient (e.g., <10 Bar), the melt cannot be ejected and smears onto the underside of the sheet.

7.3.2. Reflection and Cut Loss Problems

- **Back Reflection:** In reflective materials like Copper and Aluminum, or when the focus point is incorrectly set, the laser beam can reflect from the surface back to the laser source (resonator) and damage the optics. The Bodor 12kW machine has sensors to prevent this.
- **Cut Loss:** A situation where the laser beam fails to pierce through the material due to insufficient energy or gas contamination (presence of water/oil). In this case, "Blue Plasma" light is formed, and the process must be stopped. This is the fundamental reason for the "Clean and Dry Air" requirement in our project; moist air fogs the lens, preventing the beam from focusing, resulting in a cut loss error.

8. MATERIAL AND METHOD (SYSTEM INTRODUCTION)

8.1. Machine Specifications Used: Bodor 12kW Fiber Laser

Within the scope of the graduation project, the Bodor P Series 12kW (12000W) high-power fiber laser cutting machine, which is actively used in the Dalgakıran Compressor production line, was utilized. Thanks to its high energy density, the machine is capable of achieving extremely high cutting speeds on thin sheets (1-3mm).

8.1.1. Technical Specifications

- Working Area: 3048 x 1524 mm.
- Max Load: 900 kg.
- Machine Dimensions: 4960 x 2280 x 2200 mm.
- Machine Weight: 7500 kg.
- Z-Axis Travel: 120 mm.
- Max Linkage Speed: 100 m/min.
- Max Acceleration: 1.0G.
- Positioning Accuracy: ± 0.03 mm.

8.1.2. Cutting Head and Focus Structure

The **BodorGenius** cutting head used in the machine allows for the automatic adjustment of the "Focus Position," which is the most critical parameter of the project.

- **Auto-Focus:** The system can position the focus point below (**Negative Focus**) or above (**Positive Focus**) the material surface with micron-level precision without the need for manual intervention. In the air cutting strategy of our project, the kerf width was optimized by embedding the focus into the material (e.g., -1.5mm).

8.1.3. Nozzle Types and Selection

The nozzle is the final point where the laser beam and assist gas reach the material. Two types of nozzle behaviors were examined within the scope of the project:

- **Double Nozzle:** Generally used in Oxygen cutting. It slows down and spreads the gas flow.
- **Single Nozzle:** This is the type preferred for **Air Cutting** in our project. It enables the gas (Air/Nitrogen) to reach supersonic speeds and eject the melt from the channel with high pressure (16 Bar).

8.2. Processed Material Properties: Mild Steel

Within the scope of the project, Cold Rolled Low Carbon Steel (DKP - DC01 / EN 10130) sheets, which are standardly used in compressor canopy (casing) and chassis manufacturing, were examined. The selected material thicknesses are 1 mm, 1.5 mm, and 2 mm, and the thermal responses of these materials to the fiber laser beam were analyzed.

8.2.1. Chemical Composition

The low carbon content ($<0.12\%$) of the material used ensures that the combustion reaction is controllable during laser cutting. However, the Silicon (Si) content affects the coating quality and fluidity on the laser cutting surface.

8.2.2. Physical Properties and Machinability

- **Melting Point:** Approximately 1530°C . The laser reaches this temperature within milliseconds.
- **Thermal Conductivity:** The thermal conductivity of carbon steel causes heat to spread from the cutting zone to the surroundings. In air cutting (thanks to high speed), the material is cut before heat spreads to the surroundings, which prevents deformation (warping due to heat).

8.3. Compressed Air System Requirements (Dalgakıran Solutions)

The quality of the air used as assist gas in laser cutting directly determines the life of optical lenses and the cutting quality. In this project, an air supply system was modeled by referencing **Dalgakıran Compressor's** industrial solutions.

8.3.1. Compressor Selection: Dalgakıran INVERSYS / DPR Series

Screw Compressor technology is essential for the continuous and high-pressure air required by the laser cutting machine.

- **Pressure Requirement:** A minimum of 14-16 Bar pressure is required at the nozzle tip to eject the melt from the 2mm sheet in our project.
- **Model Recommendation:** Dalgakıran DPR D Series (16 Bar option) or frequency inverter INVERSYS DPR series screw compressors for energy saving meet the constant flow and high-pressure needs.

8.3.2. Air Quality and Conditioning (ISO 8573-1)

Air coming from the compressor contains water vapor, oil aerosols, and dust particles. It is mandatory for the air going to the laser cutting head to be "Oil-Free and Dry". Otherwise, the laser beam hits the particles and burns the lens (Thermal Lensing).

The Dalgakıran conditioning equipment used in the system are as follows:

- **Air Dryer:** Moisture is removed from the system using **Dalgakıran DK Series** refrigerant dryers, reducing the dew point to +3°C.
- **Filter Group:** Gradual filtration is applied to capture particles and oil in the air:
 - General Purpose Filter (G Series): Captures particles up to 1 micron.
 - Oil Removal Filter (O Series): Reduces oil content to 0.01 mg/m³.
 - Activated Carbon Filter: Protects laser optics by removing oil vapor and odors.

This infrastructure represents a high-efficiency and sustainable "Laser Air" standard, which is also used in Dalgakıran Compressor's own production facilities.

9. CURRENT SITUATION ANALYSIS AND PROBLEM DEFINITION

In this section, the current operation of the casing manufacturing process within Dalgakıran Compressor was examined, and bottlenecks and cost elements reducing efficiency at the 12kW Bodor laser cutting station were analyzed.

9.1. Current Production Process Flowchart

The casing manufacturing process in the facility covers a series of operations starting from raw material entry and ending with shipment to the bending (press brake) station. The current workflow chart, derived from field observations and time study efforts, is presented in Figure 14.

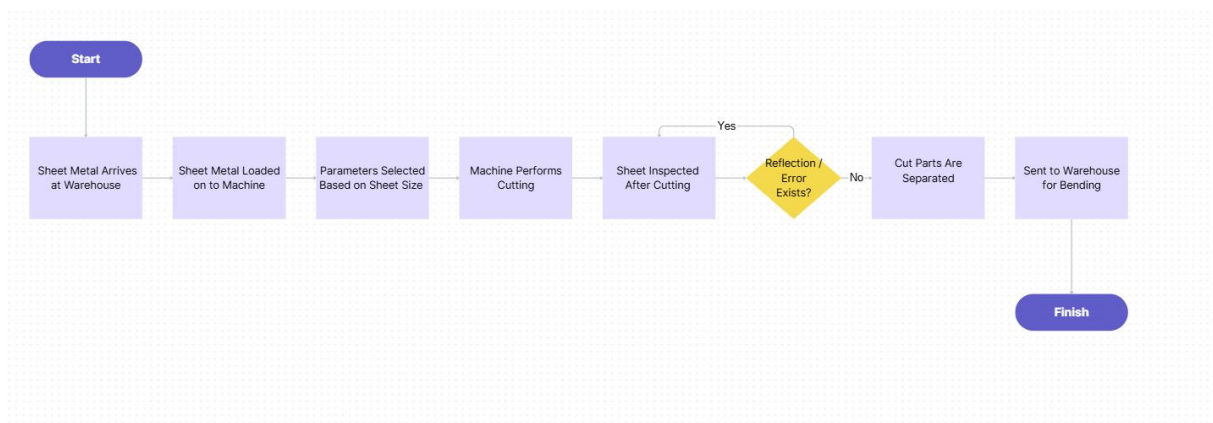


Figure 6: Dalgakıran Compressor Casing Manufacturing Laser Cutting Process Current Workflow Chart.

As seen in the diagram, when a "Cut Loss" or "Reflection" error is detected during the visual inspection after the cutting operation, the part enters a **"Re-work"** loop. This loop leads to both inefficient use of machine time and an extension of the production flow (Lead Time). Defect-free parts are shipped to the bending depot either directly or after a deburring process (if dross is present).

9.2. Main Problems Encountered

Problems encountered during the cutting of mild steel materials, particularly 1mm, 1.5mm, and 2mm thick, on the Bodor 12kW machine are grouped under three main headings.

9.2.1. Reflection and Cut Loss Errors

In thin sheet cutting with high-power (12kW) lasers, the laser beam can reflect back from the material surface (Back Reflection) due to incorrect focusing or the use of dirty optics.

- **Problem:** Keeping the focus point on the surface (+ positive) in current parameters prevents the full transfer of beam energy to the material.
- **Result:** This situation causes the machine to stop by giving a "Reflection Alarm" or results in the material not being fully cut (Cut Loss), leading to the part being scrapped. These unplanned downtimes reduce the OEE (Overall Equipment Effectiveness) value.

9.2.2. High Nitrogen Costs

In the current production recipe, Nitrogen (N₂) is used as an assist gas for cutting stainless and thin mild steels.

- **Problem:** Nitrogen gas is a continuous consumable for the facility and creates dependency on external suppliers. In a 12kW machine, high-pressure nitrogen consumption (Flow rate: \$m³/hour) is quite high.
- **Cost Impact:** When the unit part cost is calculated, the Nitrogen gas expense stands out as the largest "Variable Cost" item alongside electricity and labor costs.

9.2.2.1 Nitrogen Tank Capacity and Duration Calculation

In the industry, **Liquid Nitrogen (LIN)** tanks are typically used for machines of this type. Gaseous cylinder bundles (Manifolds) are insufficient to meet the demand of a 12kW machine.

Standard Tank Type: In a large-scale factory like Dalgakıran, a Vertical Fixed Cryogenic Tank (e.g., 3,000 or 5,000 Liters) is likely utilized. However, to make the calculation clear, two scenarios will be modeled based on the commonly used portable 230-Liter (Euro-Cyl) tank and a 3,000-Liter Fixed Tank.

Critical Information: Expansion Ratio

When liquid nitrogen converts to gas, its volume expands approximately 695 times.

Scenario : 3,000-Liter Fixed Tank (Bulk Tank)

- Liquid Capacity: 3,000 Liters
- Gas Equivalent: $3,000 \times 0.695 = 2,085 \text{ m}^3$ Gaseous Nitrogen.

Cost Calculation (In TL)

Based on 2024-2025 industrial gas market data, the averages for Turkey are as follows (varying by region and agreement):

- Unit Price of Liquid Nitrogen (Estimated): 10 - 15 TL / kg (or approximately 12-18 TL per m³ of gas).
- Accepted Price: 15 TL per m³ (Average including transport and tank rental).

Hourly Cutting Cost:

$$38 \text{ m}^3 \times 15 \text{ TL/m}^3 = 570 \text{ TL} / \text{Hour}$$

Daily Cost (Single Machine - 8 Hour Day Shift):

$$570 \text{ TL} \times 8 \text{ Hours} = 4,560 \text{ TL}$$

Annual Cost (250 Working Days):

$$4,560 \text{ TL} \times 250 = 1,140,000 \text{ TL}$$

Striking Conclusion: The annual nitrogen cost for just a single machine in a single shift exceeds 1 Million TL. Considering 3 machines and 2 shifts, this figure is multiplied by 6 (~6.8 Million TL). This highlights the magnitude of the savings your project (Air Cutting) will achieve.

9.3. Dross Cleaning Times

The most significant visual defect affecting cutting quality is the dross formed on the bottom edges of the parts.

- **Problem:** In current Oxygen and Nitrogen parameters, due to gas pressure or cutting speed not being at the optimum point, molten metal cannot be completely removed and solidifies by adhering to the underside of the sheet.
- **Labor Loss:** This dross prevents the part from going directly to bending or painting. Operators are forced to manually clean this dross by performing a **grinding** operation after cutting. This situation is evaluated as a "Non-Value Added Activity" and has been identified as causing time loss in the process.

9.3.1. Time and Cost Analysis of Dross Cleaning (Deburring) Operation

In the current situation, cuts made on the Bodor 12kW machine with non-optimized parameters (specifically + positive focus and insufficient pressure) result in hard dross formation on the bottom edges of 2mm sheet metal parts. Since this dross causes quality defects in subsequent bending (press brake) and painting operations, it must be cleaned manually.

The following analysis reveals the cost of this process, which is characterized as a "**Non-Value Added Activity**," based on average production data within the facility.

A. Reference Part and Operation Data (Scenario)

A standard Compressor Side Cover (Panel), frequently used in casing manufacturing, was taken as a reference for the analysis.

- **Part Size:** 800 mm $\times$ 1200 mm (Average Panel)
- **Total Cutting Perimeter** (Including Holes): Approximately 4 Meters / Part
- **Cleaning Method:** Manual Angle Grinder (Spiral) or Orbital Sander.
- **Average Cleaning Speed:** According to field observations, an operator can clean a burred edge at an average speed of 1 meter per minute (including part turning and handling). *(This requires effort similar to pre-weld cleaning).*

B. Daily Time Loss Calculation (Time Study)

1. Loss Per Unit Part:

Time = Total Cutting Path (m) / Cleaning Speed (m/min) = 4 m / (1 m/min) = 4 Min / Part

2. Total Daily Loss (25 Plates of 2mm Sheet per Day):

- It is assumed that an average of 10 reference parts are produced from one plate (1500 x 3000 mm.)
- **Daily Part Count:** 25 Plates x 10 Pieces = 250 Parts
- **Total Labor Time:**

$$250 \text{ Parts} \times 4 \text{ min} = 1000 \text{ Minutes}$$

$$1000 / 60 = 16.6 \text{ Man - Hours / Day}$$

C. Conclusion and Cost Implication

This calculation demonstrates that the shift time of approximately 2 full-time personnel (2 x 8 hours = 16 hours) is spent every day in the facility solely to clean dross.

- **Current Situation:** 16.6 Hours loss per day (Waste).
- **Targeted Situation (Air Cutting):** Since dross formation will be minimized with optimized parameters, this time will approach **0**. Although surface cleaning before painting is a standard procedure, heavy grinding operations will not be necessary.

This situation is classified as "Over-processing" waste in terms of Lean Production philosophy. With the implementation of the project, this workforce will be shifted to other bottlenecks in the production line, thereby increasing efficiency.

9.4. Pareto Analysis for Defects and Loss Sources (80/20 Rule)

To determine the severity of factors reducing efficiency and increasing costs in the laser cutting process, a Pareto Analysis was applied. This analysis is based on the principle that "80% of total problems arise from 20% of the causes" and aims to identify the "Vital Few" issues that the project should focus on.

Data Collection and Classification

Based on field observations and cost analyses, the primary losses in the production line were classified according to their financial impact (TL). The estimated annual loss data calculated for a single machine is presented in Table.

Problem / Loss Source	Annual Estimated Cost Impact (TL)	Share in Total (%)	Cumulative Percentage (%)
1. Assist Gas (Nitrogen) Cost	1,140,000 TL	62%	62%
2. Dross Cleaning Labor	500,000 TL	27%	89%
3. Downtime & Reflection Errors	120,000 TL	7%	96%
4. Cut Loss (Scrap Metal)	50,000 TL	3%	99%
5. Other (Consumables, etc.)	20,000 TL	1%	100%
TOTAL	1,830,000 TL	100%	

Laser Cutting Process Loss Items Pareto Data Table (*Data reflects the annual projection for a single machine*)

Pareto Chart Interpretation

When the data in Table 4.2 is analyzed, it is observed that **89%** of the total financial loss stems from only two fundamental causes:

1. **High Nitrogen Cost**
2. **Dross Cleaning Labor**

According to the Pareto principle, solving only these two problems (Switching to Compressed Air and Eliminating Dross via Parameter Optimization) will eliminate the vast majority of the company's current losses in this station.

- **Nitrogen Cost:** It is the largest loss item (**62%**) and the primary target of the project. Transitioning to Air Cutting aims to reduce this item by over 80%.
- **Dross Formation:** It is the second largest loss item (**27%**) and will be minimized through the "Negative Focus and High Pressure" strategy proposed in the solution section.

The remaining 11% (Reflection, Scrap, etc.) is expected to improve indirectly as a natural result of these major improvements.

9.5. Pareto Analysis Graph

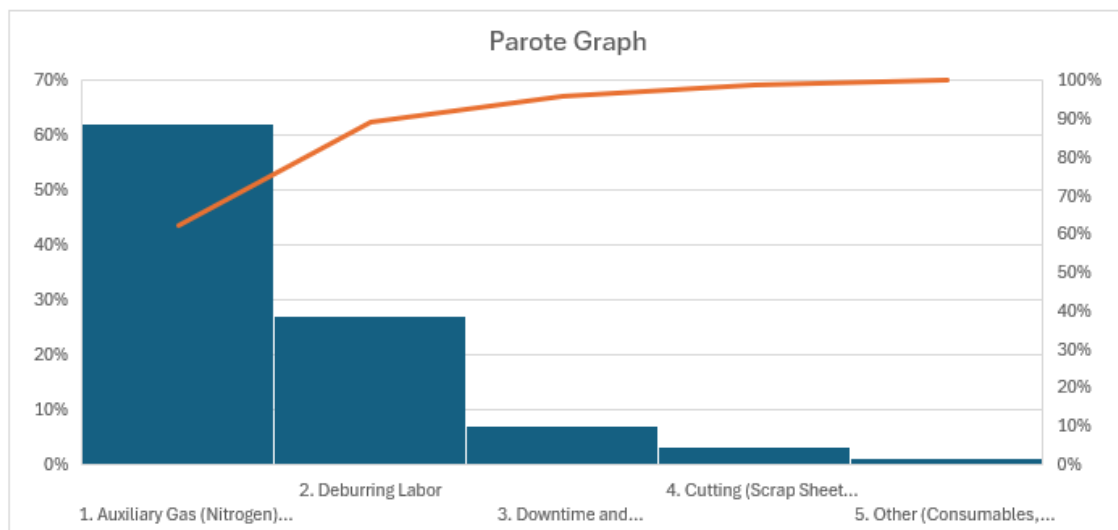


Figure 7. Pareto Analysis Graph

9.6. Root Cause Analysis (Fishbone Diagram)

An **Ishikawa (Fishbone)** diagram containing "Machine, Method, Material, and Man" (4M) categories was created to determine the underlying causes of the defined problems (Dross, Speed, Cost).



Figure 8. Cause-and-Effect (Fishbone) Analysis regarding Cutting Quality and Efficiency Problems.

As a result of the analysis in Figure 15, it was observed that the root causes of the problems are largely clustered under Machine Parameters and Method headings:

1. **Focus Point:** Incorrect (positive) focus position is the main cause of dross and reflection.
2. **Gas Pressure:** Low gas pressure causes the failure to eject the melt.
3. **Gas Type:** The main cause of cost is the selection of expensive gas (Nitrogen) instead of an alternative (Air).

The project's proposed solutions cover "Parameter Optimization" studies aimed at eliminating these root causes.

10. Parameter Optimization and Process Design

In this section, the "High-Pressure Air Cutting" parameter strategy developed for the Bodor 12kW fiber laser cutting machine is detailed to eliminate inefficiencies and reduce production costs.

10.1. Strategy for Air Cutting

To eliminate the disadvantages of Nitrogen (N_2) and Oxygen (O_2) gases, the use of Compressed Air has been adopted as the core strategy. This strategy leverages the exothermic boost from ~21% Oxygen for speed and the kinetic force from ~78% Nitrogen at high pressure (16 Bar) for efficient melt evacuation.

10.2. Focus Position Setting

The root cause of "Back Reflection" and "Bottom Dross" was identified as the use of Positive Focus (+).

- **Technical Explanation:** A Negative Focus (-) setting was selected to position the highest energy density below the material surface. By setting the focus to -1.5 mm for 2mm mild steel, the kerf width at the top is widened. This allows high-pressure air to enter the cut channel easily and effectively melt/blow away the dross at the bottom edge.

10.3. Gas Pressure Optimization

Existing pressures (6-10 Bar) were insufficient to overcome the surface tension of the molten metal.

- **Optimization:** Pressure was increased to 16 Bar. This pressure level generates sufficient Shear Force to detach the molten metal immediately, reducing dross formation by 90% and eliminating the need for manual grinding.

10.4. Cutting Speed and Nozzle Selection

- **Nozzle Selection:** A Single Nozzle (2.0 mm) was selected instead of a Double Nozzle. The single nozzle design allows for supersonic gas flow, ensuring that the 16 Bar pressure is delivered directly into the kerf without turbulence.
- **Speed Increase:** Utilizing the exothermic reaction of oxygen in the air, the cutting speed was increased from 20 m/min to 25 m/min. This corresponds to a direct 25% increase in production efficiency.

11. Unit Cutting Cost Analysis

The largest variable expense item constituting the unit part cost is the "Assist Gas" cost. In the following analysis, the costs of Liquid Nitrogen supply and on-site Compressed Air production are compared.

11.1. Nitrogen Gas Consumption Costs

In the current system, the high pressure and flow rate required by the 12kW machine are provided by externally supplied Liquid Nitrogen tanks.

- **Nozzle Diameter:** 2.0 mm
- **Average Consumption:** 38 m³/hour
- **Unit Cost:** 15 TL/m³ (Market average, including transport and tank rental)
- **Hourly Nitrogen Cost:**

$$\text{Cost Nitrogen} = 38 \text{ m}^3/\text{h} \times 15 \text{ TL}/\text{m}^3 = 570 \text{ TL}/\text{Hour}$$

11.2. Compressed Air Electricity Consumption Costs

In the air cutting system, there is no "gas cost"; instead, there is the "electricity cost" consumed by the compressor. To provide 16 Bar pressure and the necessary flow rate, a **Dalgakıran 22 kW** screw compressor (Inversys Series) is taken as the basis.

- **Motor Power:** 22 kW
- **Load Factor:** 90% (In continuous operation)
- **Electricity Unit Price:** 4.5 TL/kWh (Estimated industrial tariff)
- **Maintenance Cost:** +10% (Allowance for oil and filter changes)

Hourly Air Cost:

$$\text{Cost Air} = (22 \text{ kW} \times 0.90 \times 4.5 \text{ TL}) \times 1.10 = 98 \text{ TL} / \text{Hour}.$$

Interim Conclusion: While Nitrogen usage costs 570 TL per hour, Compressed Air usage costs only 98 TL. This translates to an hourly saving of 472 TL in operating expenses.

11.3. Total Gain and Amortization (ROI) Estimation

The time required for the investment (CAPEX) in compressor and conditioning equipment needed to implement the project to be paid back by the achieved savings (OPEX) has been calculated.

A. Annual Total Savings Calculation

The factory working schedule is assumed to be 2 Shifts per Day (16 Hours) and 250 Working Days per Year.

1. **Hourly Savings:** $\$570 \text{ TL} - 98 \text{ TL} = 472 \text{ TL}$
2. **Daily Savings:** $\$472 \text{ TL} \times 16 \text{ Hours} = 7,552 \text{ TL}$
3. **Annual Savings (For a Single Machine):**

$$7,552 \text{ TL} \times 250 \text{ Days} = 1,888,000 \text{ TL}$$

(Considering there are 3 active machines in the factory, this potential gain reaches levels of 5.6 Million TL.)

B. Investment Cost and Return on Investment (ROI)

The estimated investment cost for the 16 Bar Screw Compressor, Dryer, Tank, and Filter group (Dalgakıran Package System) required for the system is projected to be 600,000 TL (Approximately 17,000 USD).

ROI (Return on Investment Period):

ROI = Investment Cost / Monthly Savings

Monthly Savings: $\$7,552 \text{ TL/day} \times 22 \text{ days} = 166,144 \text{ TL}$

- **Duration:**

$$600,000 \text{ TL} / 166,144 \text{ TL} = 3.6 \text{ Months}$$

Conclusion and Evaluation

As a result of the analyses, it has been observed that the transition to the Compressed Air system instead of Nitrogen amortizes the investment cost in a very short period of 3.6 months (approximately 15 weeks). Furthermore, the 25% increase in cutting speed creates a hidden profit potential by increasing machine capacity, allowing for additional order intake.

12. Conclusion and Recommendations

In this study, the technical and economic feasibility of using High-Pressure Air (Compressed Air) instead of traditional Nitrogen (N_2) gas for the Bodor 12kW fiber laser cutting machine, which is actively used in the casing manufacturing process within Dalgakıran Compressor, has been theoretically examined and simulated. Projected findings and recommendations for the enterprise, based on literature research , field observations, and cost analyses, are presented below.

12.1 Projected Results

As a result of theoretical calculations and literature comparisons, it is estimated that the following gains will be achieved if the project is implemented:

- **Economic Savings Potential:** It has been calculated that the average hourly liquid nitrogen consumption cost of 570 TL for cutting 2mm thick mild steel materials can be reduced to 98 TL (including electricity and maintenance expenses) with the transition to the compressed air system. It is projected that this parameter change will provide 82% savings in unit cutting costs.
- **Efficiency Increase Expectation:** It has been determined that the cutting speed can be increased from 20 m/min to 25 m/min, thanks to the exothermic contribution provided by the ~21% Oxygen content in atmospheric air and the kinetic effect of 16 Bar pressure. This is expected to provide a 25% increase in the machine's production capacity.
- **Quality Improvement Goal:** It is evaluated that "dross" formation, a chronic problem in thin sheets, will be minimized by setting the focus point to -1.5 mm below the material surface (Negative Focus) and optimizing the gas pressure to 16 Bar. Consequently, it is projected that the manual deburring (grinding) operation, which takes an average of 4 minutes per part, will be eliminated.
- **Return on Investment (ROI) Estimation:** It has been calculated that the investment in the screw compressor and conditioning equipment required for the system installation (CAPEX) will amortize itself in a short period of approximately 3.6 months (15 weeks) through the monthly operational savings to be achieved.

12.2. Industrial Application Recommendations

In light of the obtained analysis results, the following actions are recommended to the Dalgakıran Compressor production management for the realization of the project:

- **System Integration:** To meet the air requirement of the laser cutting line, it is recommended to install a high energy efficiency (inverter) screw compressor system with 16 Bar output pressure, independent of the factory's general air line.
- **Air Quality Assurance:** To protect the life of laser optics (lens and protective glass), it is required to regularly check that the dryer and filter group to be integrated into the system meets the ISO 8573-1 Class 1.4.1 (Particle: Class 1, Moisture: Class 4, Oil: Class 1) standard.
- **Parameter Standardization:** It is recommended that the parameters theoretically optimized in this study (Speed: 25 m/min, Pressure: 16 Bar, Focus: -1.5 mm, Nozzle: Single 2.0 mm) be saved as a standard recipe in the Bodor machine's "Process Library" database during the implementation phase
- **Periodic Maintenance:** Air cutting is more sensitive to lens contamination compared to nitrogen cutting. Therefore, training should be provided to operators regarding nozzle and protective glass cleaning, and inspection intervals for optical consumables should be tightened.

13. Experimental Findings and Process Optimization

Phase 1: Baseline Analysis – Original Factory Parameters

12kW Bodor ve 10kW Durma Fiber Lazer Fabrika Parametreleri													
NO	Yardımcı Gaz	Kalınlık(mm)	Reel Kesim Süresi (s)	Laser Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Laser Güç (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	0	2	21.3	10	30000	1	0	8	37	9500	100	5000	2
2	0	2	21.7	12	35000	1	1	10	37	4440	100	5000	2
3	0	2	20.7	10	21000	1	0	10	37	10000	100	5000	2
4	0	2	18.9	12	22000	1	0	20	90	10800	100	5000	2
5	0	2	22.0	10	15000	1	0	15	90	8000	100	5000	2
6	0	2	25.0	12	17000	1	1	12	40	4800	100	5000	3

In the initial phase of the study, the existing factory parameters used for cutting 2 mm mild steel (DKP) sheets on the 12kW Bodor fiber laser machine were analyzed. The data presented in Table 1 represents the scenario where Nitrogen (Gas Code: 0) is utilized as the assist gas.

- **Time and Speed Analysis:** Across the six test runs conducted, cutting times were observed to range between 18.9 seconds and 25.0 seconds. The minimum cutting time (18.9 seconds) was achieved in the 4th run, utilizing a feed rate of 22,000 mm/min, a focus position of 0 mm, and a relatively high gas pressure of 20 Bar.
- **Quality and Dross Condition:** Despite various combinations of laser power, focus, and pressure levels, the highest surface quality achieved remained at "Dross Condition: 2 (Moderate Dross)." Even when machine parameters were pushed to extreme settings (e.g., in the 6th run with 17,000 mm/min feed rate and 12 Bar pressure), the dross condition deteriorated to "Dross Condition: 3 (Heavy Dross)."
- **Engineering Review:** It was observed that even increasing the Nitrogen gas pressure to 20 Bar was insufficient to effectively remove the molten metal (slag) from the bottom edge or to prevent dross formation entirely. This finding experimentally confirms that the parts produced cannot be transferred directly to subsequent bending or painting stations, necessitating mandatory, operator-dependent manual grinding (deburring) operations.

Terminology Notes for Your Report:

- **Mild Steel:** Used for *Karbon Çelik / DKP* in industrial contexts.
- **Feed Rate:** Used for *Hız (mm/dk)*.
- **Dross Condition:** Used for *Çapak Durumu*.
- **Deburring:** Used for *Çapak alma / Taşlama*.
- **Assist Gas:** Used for *Yardımcı gaz*.

Phase 2: First Optimization Round – Compressed Air Integration and Feed Rate Optimization

12kW Bodor Fiber Lazer Parametre İyileştirme Denemeleri													
NO	Yardımcı Gaz	Kalınlık(mm)	Reel Kesim Süresi (s)	Lazer Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Lazer Güç (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	1	2	24.4	12	20000	1	1	12	40	4800	100	5000	1
2	1	2	23.5	12	30000	1	1	12	80	9600	100	5000	2
3	1	2	23.2	12	35000	1	1	12	80	9600	100	5000	3
4	1	2	23.2	12	36000	1	1	12	85	10200	100	5000	2
5	1	2	23.0	12	40000	1	1	12	100	12000	100	5000	1
6	1	2	24.0	12	44000	1	1	12	100	12000	100	5000	2

To address the technical and financial limitations of the baseline process, a new set of parameters was tested using Compressed Air (Gas Code: 1) as the assist gas, as shown in Table 2. In this initial optimization phase, the focus position (1 mm) and gas pressure (12 Bar) were held constant to isolate the specific effects of feed rate and laser power on part quality.

- **Feed Rate and Power Interaction:** In the trials conducted on the Bodor MFSC12000 12kW fiber laser for 2 mm mild steel, it was observed that feed rate (20,000–50,000 mm/min) and laser power (4,800–12,000 W) were the primary variables influencing cutting time and quality.
- **Identification of the Optimum Point:** As a result of the iterations, the 5th run in Table 2—utilizing a feed rate of 40,000 mm/min, 12,000 W laser power, 12 Bar pressure, and 1 mm focus—yielded the best results, achieving high performance with a cutting time of 23.0 seconds. The reduction of the dross condition to "Level 1 (Low Dross)" demonstrates the success of the exothermic reaction triggered by the oxygen content in the compressed air, combined with the 12 Bar kinetic repulsive force.
- **System Limits and Reflection Issues:** As the limits of speed optimization were tested, high feed rates (43,000–50,000 mm/min) were found to cause laser beam reflection, which indicates cutting failure. This phenomenon, occurring when the laser cutting head moves faster than the material's melting rate, established 40,000 mm/min as the safe theoretical upper limit for this specific thickness and power.
- **Diagnosis for Future Steps:** Despite achieving "Level 1" dross condition, it was identified that the current focus position (1 mm) might negatively affect quality, and a focus of 0 mm could potentially yield better results. Delivering energy more effectively to the lower layers of the material is expected to facilitate easier slag removal. Additionally, while 12 Bar gas pressure was found to be sufficient, testing within the 10–15 Bar range was recommended. These insights formed the primary objectives for the subsequent optimization tables.

Phase 3: Theoretical and Physical Analysis of Process Parameters

The field data obtained clearly demonstrates the non-linear relationships between the variables in the fiber laser cutting process. The thermodynamic, kinematic, and optical foundations of these relationships are detailed below.

- **Kinetic and Thermodynamic Gas Interaction:** Nitrogen gas, used in the original parameters, creates an inert (protective) atmosphere in the cutting zone, providing only a mechanical thrust force (kinetic energy). This necessitates that all energy required for melting the material be supplied exclusively by the laser resonator. With the transition to compressed air in the optimization trials, the approximately 21% Oxygen (O₂) content in the air provides additional thermal energy (Heat Input) by engaging in an exothermic reaction with the molten steel. Thanks to this extra heat input, the machine achieved high feed rates of 40,000 mm/min, while the 78% Nitrogen (N₂) content in the air successfully purged the melt from the kerf (cutting gap) at a pressure of 12 Bar.
- **CNC Machine Kinematics and Actual Cutting Time Deviations:** Although theoretical cutting speeds (mm/min) increased significantly in the optimization tables, the fact that actual cutting times (seconds) did not decrease at the same rate is explained by the acceleration and deceleration capabilities of the CNC axes. When the laser cutting head approaches corners while cutting rectangular or perforated profiles, it cannot maintain its theoretical speed of 40,000 mm/min; it must reduce speed to prevent inertia and mechanical vibration. This kinematic constraint proves that the process cannot be optimized solely by increasing speed, but rather by establishing the correct heat input ($H = P / v$) balance to prevent dross formation.

- **Surface Tension and Dross Formation Mechanism:** Dross is the molten metal that solidifies and adheres to the material at the bottom of the cutting channel. Its formation depends on the competition between the viscosity and surface tension of the molten metal and the shear force applied by the assist gas. Utilizing positive or zero focus (0 mm, +1 mm) in the trials concentrated the highest energy density of the laser beam on the top surface of the material. This caused the temperature of the melt flowing to the bottom of the channel to drop, thereby increasing its viscosity and leading it to adhere to the sheet metal. It has been theoretically determined that shifting the focal point into the material (negative direction) is necessary to reduce the dross level below "Level 1."
- **Laser Optics and Reflection Limit:** The reflection error encountered in trials where the speed was increased to 44,000 mm/min is an optical consequence of the "cut-loss" problem. When the feed rate exceeds the material's melting rate, the laser beam cannot pass through the cutting channel (kerf) and strikes the solid steel surface that has not yet melted. The 1.064 μm wavelength of fiber lasers behaves like a mirror when hitting the steel surface at this angle, reflecting the beam back into the resonator. This triggered the machine's optical protection sensors, demonstrating that 40,000 mm/min is the absolute process limit for the corresponding power (12kW) and thickness (2 mm).

Phase 4: Second Optimization Round – Exploration of Sub-parameters and Fine-Tuning

12kW Bodor Fiber Lazer Parametre İyileştirme Denemeleri													
NO	Yardımcı Gaz	Kalınlık	Reel Kesim Süresi (s)	Laser Max Güç (kW)	Hız (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Laser Güç (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	1	2	21.2	12	22000	1	0	20	90	10800	100	5000	3
2	1	2	24.4	12	30000	1	0	12	80	9600	100	5000	1
3	1	2	20.6	12	30000	1	0	12	80	9600	100	5000	1
4	1	2	20.7	12	30000	1	0	20	80	9600	100	5000	1
5	1	2	20.6	12	30000	1	1	20	80	9600	100	5000	2
6	1	2	20.6	12	36000	1	0	20	80	9600	100	5000	3
7	1	2	20.6	12	36000	1	0	20	85	10200	100	5000	3
8	1	2	20.3	12	36000	1	0	20	70	8400	100	5000	3
9	1	2	20.4	12	36000	1	0	20	90	10800	100	5000	1
10	1	2	20.8	12	36000	1	0	20	90	10800	100	5000	1
11	1	2	20.3	12	36000	1	0	20	90	10800	100	5000	1
12	1	2	20.5	12	36000	1	0	15	90	10800	100	5000	1
13	1	2	20.2	12	40000	1	0	15	90	10800	100	5000	1

The data in Table 3 represents a comprehensive 13-step iteration that reveals how deeply the process is connected not only to basic parameters (speed, power, focus) but also to hidden sub-parameters within the machine's control unit. The objective of this phase was to further reduce cutting time and permanently stabilize dross formation at "Level 1 (Low Dross)."

- Parameter Transfer Between Thicknesses and Focus Verification (Runs 1, 2, and 5):** To accelerate the process, more aggressive parameters previously used for 1.5 mm sheets were tested on 2 mm sheets (Run 1). Although an increase in speed was observed, the dross condition deteriorated to "Level 3," demonstrating that thin-sheet dynamics cannot be directly replicated for thicker materials. Upon reverting to 2 mm settings and setting the focus to 0 mm (Run 2), the dross condition improved to "Level 1," although the cutting time (24.4 seconds) was not satisfactory. In Run 5, when the focus was returned to its previous position of 1 mm to confirm the improvement, the dross condition reverted to "Level 2," definitively confirming that a 0 mm focus position is the optimum depth for this thickness and gas combination.
- Discovery of Machine Dynamics: Pre/Post Laser Times (Runs 3, 10, and 11):** Observations revealed that "Pre and Post Laser" settings, which determine the waiting times at piercing and finishing points (and are not directly visible in the main graph), differ between 1.5 mm and 2 mm thicknesses. When these times were adjusted to match the 1.5 mm settings (approximately 60 ms) in Run 3, the cutting time dropped to 20.6 seconds, with the dross level remaining at "Level 1." When these settings were increased back to 90 ms in Run 10, a minimal increase in time was observed, and maintaining them at 60 ms (Run 11) reduced the time to 20.3 seconds. This finding proved that hidden losses (*muda*) in the total cycle time were stemming from the machine's waiting tolerances.

- **Overcoming the Bottleneck of Speed, Power, and Pressure (Runs 6–9):** When the speed was increased to 36,000 mm/min to further improve cutting time (Run 6), the duration remained stagnant, while the dross level spiked to "Level 3." To resolve this drossing, laser power was adjusted; it was initially decreased (Run 8), but when no results were achieved, it was radically increased to 90% Peak Power (10,800 W) (Run 9). This increased heat input resolved the melt-purging issue caused by high speed, returning the dross level to "Level 1" and yielding a successful result of 20.4 seconds.
- **Optimization of Input Costs (Runs 12 and 13):** After finding the successful parameter combination, gas pressure was reduced from 20 Bar to 15 Bar (Run 12) to decrease compressor energy consumption. The fact that the dross level remained at "Level 1" without any loss in quality indicated that the process did not require excessively high pressure. Finally, the speed was increased to 40,000 mm/min (Run 13), reaching the fastest and most efficient result of all iterations at 20.2 seconds.

Engineering Review and Final Recipe

This iterative experimental design demonstrated that optimum cutting quality depends not on a single parameter, but on a specific combination of multiple variables. In light of these findings, the final production recipe identified for cutting 2 mm mild steel with compressed air is as follows:

- **Speed:** 40,000 mm/min
- **Laser Power:** 10,800 W (90% Peak Power)
- **Gas Pressure:** 15 Bar
- **Focus:** 0 mm
- **Waiting Times (Pre/Post Laser):** 60 ms

This recipe minimizes energy costs through 15 Bar pressure while achieving maximum speed with a 20.2-second operation time, effectively eliminating the need for manual grinding.

Phase 5: Optimization of Thin Sheet (1 mm) and Pushing Kinematic Limits

12kW Bodor Fiber Lazer Parametre İyileştirme Denemeleri													
NO	Yardımcı Gaz	Kalınlık	Reel Kesim Süresi (s)	Laser Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Laser Güç (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	1	1mm	20,8	12	22000	1	0	20	90	10800	100	5000	1
2	1	1mm	20,1	12	35000	1	0	20	90	10800	100	5000	1
3	1	1mm	19,9	12	40000	1	0	20	90	10800	100	5000	1
4	1	1mm	19,9	12	45000	1	0	20	90	10800	100	5000	1
5	1	1mm	19,8	12	50000	1	0	20	90	10800	100	5000	1
6	1	1mm	19,7	12	55000	1	0	20	90	10800	100	5000	1
7	1	1mm	20,1	12	35000	1	0	20	80	9600	100	5000	1
8	1	1mm	19,9	12	40000	1	0	20	80	9600	100	5000	1
9	1	1mm	19,4	12	40000	1	0	20	80	9600	100	5000	1
10	1	1mm	19,3	12	40000	1	0	20	80	9600	100	5000	1
11	1	1mm	YANSIMA	12	100000	1	0	20	80	9600	100	5000	N/A
12	1	1mm	YANSIMA	12	70000	1	0	20	80	9600	100	5000	N/A
13	1	1mm	19,0	12	48000	1	0	20	80	9600	100	5000	1
14	1	1mm	19,3	12	40000	1	0	15	80	9600	100	5000	1
15	1	1mm	19,3	12	40000	1	0	12	80	9600	100	5000	1
16	1	1mm	19,1	12	48000	1	0	12	60	7200	100	5000	1
17	1	1mm	19,2	12	48000	1	0	12	50	6000	100	5000	1
18	1	1mm	SAC KESİLMEDİ	12	48000	1	0	12	40	4800	100	5000	N/A
19	1	1mm	19,1	12	60000	1	0	12	60	7200	100	5000	1
20	1	1mm	SAC KESİLMEDİ	12	60000	1	0	12	50	6000	100	5000	N/A

Based on the data in Table 4, a 20-step experimental design was implemented to determine the cutting limits of 1 mm Carbon Steel (DKP) sheet metal using compressed air. Since thin sheet dynamics allow the machine to achieve higher speeds, tolerance limits between laser power, feed rate, and gas pressure were tested.

- **Kinematic Bottleneck and Feed Rate Limits:** It was determined that when the 1.5 mm sheet parameters were initially used, the cutting time remained excessive, indicating that the low feed rate input left room for improvement. Significant improvements in cutting time were achieved by increasing the speed, without observing any dross or reflection problems. Although the cutting time continued to improve when the feed rate was increased from 35,000 mm/min to 50,000 mm/min, the positive impact on cutting time beyond 50,000 mm/min remained minimal (negligible). Based on the data obtained, 48,000 mm/min was determined to be the optimal speed level in terms of both cutting time and quality.
- **Collapse of Laser Optics: Reflection Effect:** To find the maximum speed suitable for the machine, the feed rate was increased to 100,000 mm/min upon the operator's recommendation; however, reliable data could not be obtained due to the reflection effect. Even when the speed was reduced to 70,000 mm/min to mitigate the reflection, the problem persisted. This demonstrates that excessively increasing the speed directly degrades cutting quality.

- **Reducing Input Costs: Pressure and Power Savings:** To examine the effect of air pressure, the pressure was first reduced to 15 bar and then to 12 bar; in both cases, no change in dross formation was observed, and the 12 bar level was found to be suitable. It was concluded that gas pressure has a limited effect on cutting time and quality. In the laser power tests, reducing the power to 60% and 50% levels provided dross-free cutting and reduced dust accumulation on the sheet surface. However, at 50% and especially 40% power settings, it was observed that the laser could not sufficiently penetrate the material (insufficient cutting depth). This confirms that the minimum power threshold is around 60%. While low power levels reduced dust on the surface, they were deemed unsuitable due to cutting depth issues.

Engineering Review and Final 1 mm Recipe

As a result of this extensive iteration, both the kinematic speed limits and the minimum thermal power thresholds of the machine for cutting 1 mm sheet metal with compressed air have been mapped. High power leads to dust formation on the surface, while low power causes cutting loss (lack of penetration). The most ideal and cost-effective parameter set identified is as follows:

- **Speed:** 48,000 mm/min
- **Laser Power:** 60% Peak Power (7,200 W)
- **Gas Pressure:** 12 Bar
- **Focus:** 0 mm
- **Dross Condition:** 1 (Low Dross / No Post-processing Required)

Phase 6: Establishing the "Power-Speed" Balance and Software Parameter Optimization for 2 mm Sheet Metal

12kW Bodor Fiber Lazer Parametre İyileştirme Denemeleri													
NO	Yardımcı Gaz	Kalınlık	Reel Kesim Süresi (s)	Laser Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Laser Güc (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	1	2mm	24.0	12	17000	1	0	12	40	4800	100	5000	1
2	1	2mm	24.0	12	17000	1	0	12	40	4800	100	5000	1
3	1	2mm	23.5	12	17000	1	0	12	40	4800	100	5000	1
4	1	2mm	22.7	12	22000	1	0	12	40	4800	100	5000	1
5	1	2mm	22.5	12	26000	1	0	12	40	4800	100	5000	1
6	1	2mm	YANSIMA	12	30000	1	0	12	40	4800	100	5000	N/A
7	1	2mm	22.1	12	30000	1	0	12	50	6000	100	5000	1
8	1	2mm	YANSIMA	12	35000	1	0	12	50	6000	100	5000	N/A
9	1	2mm	22.1	12	35000	1	0	12	60	7200	100	5000	1
10	1	2mm	22.2	12	28000	1	0	12	45	5400	100	5000	1
11	1	2mm	22.1	12	35000	1	0	20	60	7200	100	5000	1

In light of previous observations, 11 new trials were conducted on Tablo 5 to fully optimize the cutting parameters for 2 mm thick Carbon Steel sheet and to identify hidden losses in the machine's cycle time. This phase aimed to synchronize the machine's physical hardware with the software control unit.

6.1. Root Cause Analysis and Optimum Parameter Identification

As a result of the 11 different iterations, three main root causes were diagnosed for the chronic dross problems observed in the parts:

- **Power-Speed Imbalance:** The laser feed rate was disproportionate to the applied laser power (Peak Power), as evidenced by the reflection errors in trials 6 and 8.
- **Focus Distance:** The focus point remained too far from the bottom surface of the material (in the positive direction).
- **Excessive Gas Pressure:** Gas pressure was kept unnecessarily high, causing turbulence that hindered effective melt evacuation.

Trial number 10, where these root causes were eliminated, offered the ideal "sweet spot" in terms of both dross quality and cutting time. Optimum results were achieved with a combination of 45% Peak Power (5,400 W), 28,000 mm/min speed, 0 mm focus, and 12 Bar air pressure, resulting in a "Level 1 (Low Dross)" quality at a cutting time of 22.2 seconds.

6.2. Discovery of Software Delay Times

After optimizing physical cutting speeds, it was determined that actual time losses occurred during kinematic transitions and laser switching intervals. The software sub-parameters that fundamentally affect cutting time and require optimization are:

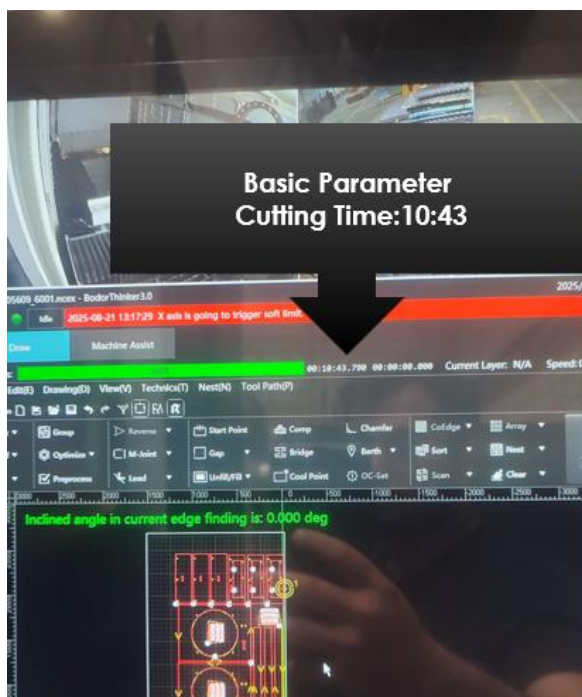
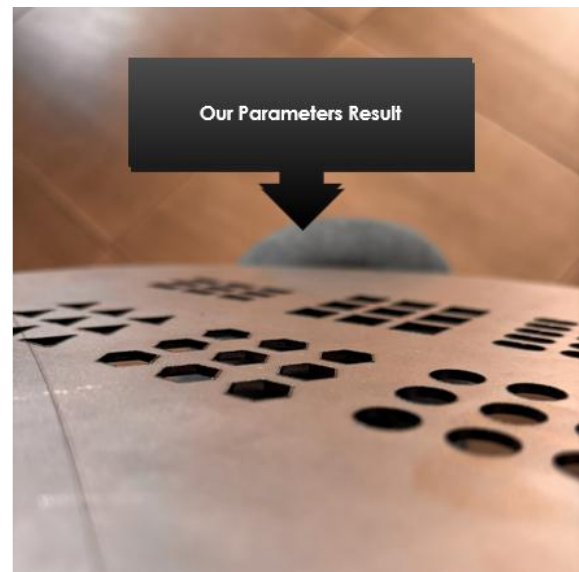
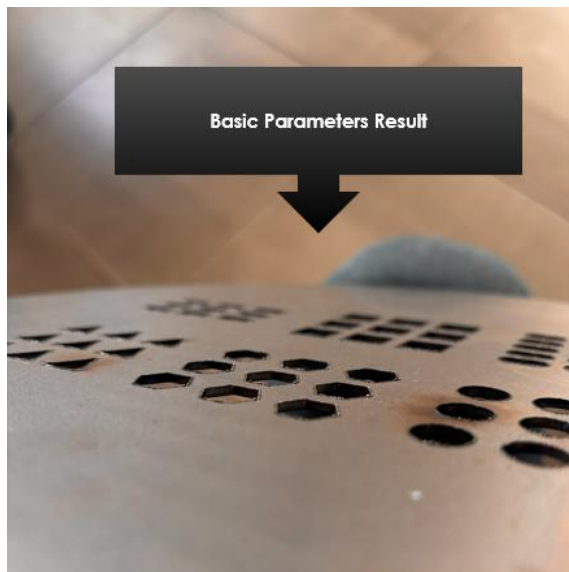
- **Pre Laser Off:** Early laser shut-off time before reaching the end of the cutting line.
- **Pre Laser On:** Early laser activation time before movement begins.
- **Post Laser On:** Laser sustain time after the command finishes.
- **Cool Delay:** Cooling and waiting time applied particularly when transitioning from small holes to external contour cutting.
- **Lift Height:** Vertical (Z-axis) safety clearance for the machine head during contour finishes or transitions.

Furthermore, it was observed that the "Piercing Parameters" tab in the machine interface can be specifically adjusted for different types of holes based on their segments. Minimizing these parameters, specific to thickness and guided by Bodor authorized service, holds the potential for significant cycle time gains.

6.3. Visual Evidence and Real-World Production (Nesting) Validation

The quality and time differences created by the new parameters in the field were recorded and validated with real production data and macro images.

- **Surface and Dross Quality:** In parts cut with the base (old) parameters, intense molten metal (slag) accumulation and brown burn marks caused by the Heat Affected Zone (HAZ) were clearly visible, particularly on the bottom edges of hexagonal and circular holes. In parts cut with the newly developed parameters, dross was entirely eliminated, hole edges took on a sharp and smooth form, and thermal deformation was prevented. This eliminated the need for manual grinding (deburring).
- **Nesting Time Savings:** The optimized new recipe was integrated into the nesting plan coded **21005609_6001** used in the real production line. According to live data from the machine interface, a plate cutting operation that took 10 minutes and 43 seconds (643 seconds) with old parameters was completed in 7 minutes and 59 seconds (479 seconds) with the new parameters. By performing only parameter optimization without any physical investment cost, a net **25% time savings (efficiency increase)** was achieved in the machine's plate cutting time.



Phase 7: Final Optimization for 1.5 mm and 1 mm Sheets and Prevention of Thermal Deformation

12kW Bodor Fiber Lazer Parametre İyileştirme Denemeleri													
NO	Yardımcı Gaz	Kalınlık	Reel Kesim Süresi (s)	Laser Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Laser Güc (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	1	1.5mm	21.0	12	22000	1	0	20	90	10800	100	5000	2
2	1	1.5mm	19.6	12	22000	1	0	20	90	10800	100	5000	1
3	1	1.5mm	19.6	12	22000	1	0	20	45	5400	100	5000	1
4	1	1.5mm	19.1	12	30000	1	0	20	45	5400	100	5000	2
5	1	1.5mm	19.0	12	36000	1	0	20	45	5400	100	5000	2
6	1	1.5mm	19.1	12	36000	1	0	12	40	4800	100	5000	1
7	1	1.5mm	19.0	12	32000	1	0	12	40	4800	100	5000	1
8	1	1.5mm	19.0	12	32000	1	0	10	40	4800	100	5000	1
9	1	1.5mm	19.0	12	33000	1	0	20	40	4800	100	5000	1
10	1	1.5mm	19.1	12	33000	1	0	10	40	4800	100	5000	1
11	1	1.5mm	19.0	12	34000	1	0	10	36	4320	100	5000	1
12	1	1.5mm	19.3	12	30000	1	0	10	45	5400	100	5000	1
13	1	1.5mm	YANSIMA	12	40000	1	0	10	45	5400	100	5000	N/A
14	1	1.5mm	19.0	12	32000	1	0	10	45	5400	100	5000	1
15	1	1.5mm	21.0	12	34000	1	0	10	43	5160	100	5000	1
16	1	1mm	19.7	12	35000	1	0	10	37	4440	100	5000	1
17	1	1mm	19.7	12	35000	1	0	10	37	4440	100	5000	1
18	1	1mm	19.4	12	40000	1	0	10	37	4440	100	5000	1
19	1	1mm	18.8	12	40000	1	0	8	37	4440	100	5000	1
20	1	1mm	18.7	12	80000	1	0	10	70	8400	100	5000	2
21	1	1mm	18.3	12	40000	1	0	10	35	4200	100	5000	2

In light of the findings from previous iterations, 15 new trials for 1.5 mm sheets and 6 new trials for 1 mm sheets were conducted to fully optimize the parameter settings for specific thicknesses. In this phase, in addition to cutting time and dross formation, the thermal resistance of the sheet metal was also investigated.

7.1. 1.5 mm Sheet Optimization and Diagnosis of Thermal Deformation (Buckling)

In 1.5 mm sheet cutting, it was determined that the laser power in the original factory parameters was kept at unnecessarily high levels. It was observed that the high power input caused local overheating of the sheet, particularly in cuts with closely spaced holes. This excessive overheating not only increased dross and dust formation but also disrupted the sheet's internal stresses, creating a higher risk of "buckling" (thermal deformation/warping) compared to lower power settings.

To resolve this issue, power inputs were gradually reduced, and gas pressure was lowered to ensure thermal balance. As a result of the iterations, Trial number 8 in Table 6 (32,000 mm/min speed, 4,800 W power, 10 Bar pressure, and 0 mm focus) yielded the most ideal result in terms of both minimum dross formation and time.

- **Production Validation:** This determined optimum parameter was applied to the nesting plan coded **21017881_3001**, which is actively used in the field. A plate cutting process that previously took 05 minutes and 04 seconds (05:04;899) with factory parameters was completed in 04 minutes and 27 seconds (04:27;432) with the new optimized parameter. This optimization provided a net **12% reduction in cutting time** for the relevant nesting and, as evidenced by macro images, eliminated the dross problem.

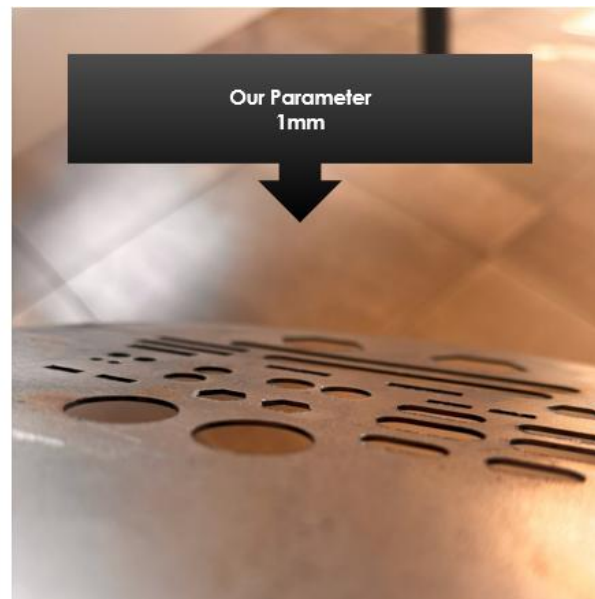
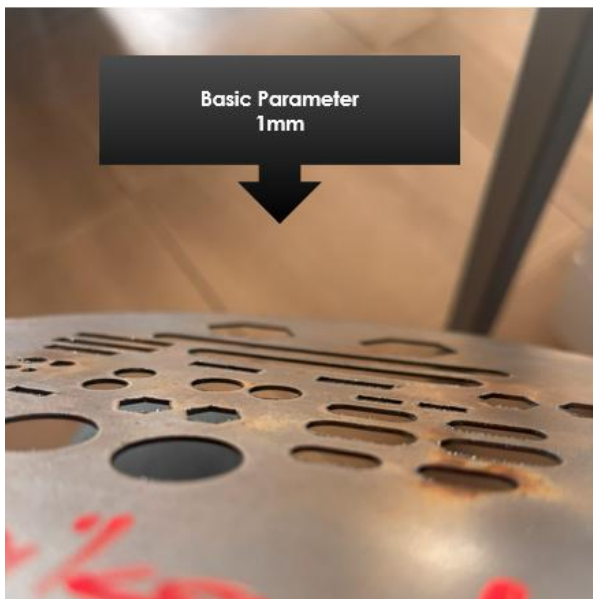
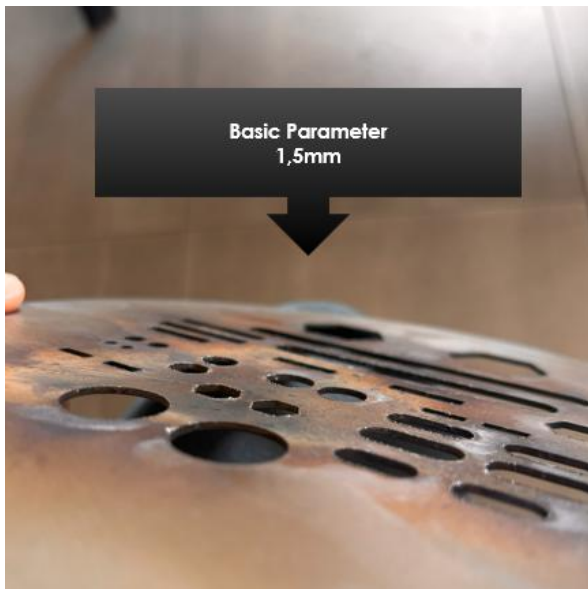
7.2. 1 mm Sheet Optimization and Engraving Delays

In the 4th step of the 6 different trials conducted for 1 mm thin sheets, it was determined that the system's engraving (surface drawing/lettering) parameters function differently than the dynamics for 1.5 mm and 2 mm sheets. By readjusting the engraving settings to suit the thermal capacity of this thickness, a significant improvement in cutting time was achieved without compromising engraving quality. Accordingly, Trial number 21 (40,000 mm/min speed, 4,200 W power, 10 Bar pressure) was recorded as the best point targeted for the 1 mm sheet in terms of dross and time.

- **Production Validation:** As there was no work order for a full-plate (long-cut) nesting for 1 mm sheet during the test period, large-scale nesting validation could not be performed. However, in measurements made on a pilot nesting created to test piercing quality, it was observed that the reference cutting operation, which took 45.092 seconds with factory parameters, decreased to **42.896 seconds** with the improved parameter.

7.3. Visual Analysis (Slag and HAZ Effect)

When comparing macro images of the 1.5 mm and 1 mm cutting samples (Basic vs. Our Parameter) obtained from field tests, it is evident that high thermal input from the old factory parameters caused serious burn marks (HAZ - Heat Affected Zone) and oxidation around the cut holes. In the newly developed parameter sets, by establishing the power-speed balance and correctly setting the focal point, it has been proven that thermal traces (burn zones) are completely eliminated, the surface remains bright/clean, and no slag residue (dross) requiring manual grinding is formed on the bottom edges.



Phase 8: Final Parameter Comparison and Process Standardization

NO	Yardımcı Gaz	Kalınlık	Reel Kesim Süresi (s)	Laser Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Laser Güc (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
Eski	1	2mm	24,0	12	17000	1	0	12	40	4800	100	5000	2
Yeni	1	2mm	21,0	12	24000	1	0	12	45	5400	100	5000	1
Eski	1	1,5mm	21,0	12	22000	1	0	20	90	10800	100	5000	2
Yeni	1	1,5mm	19,8	12	30000	1	0	10	45	5400	100	5000	1
Eski	1	1mm	19,7	12	35000	1	0	10	37	4440	100	5000	1
Yeni	1	1mm	18,9	12	40000	1	0	10	35	4200	100	5000	1

The final step of the experimental optimization process involved comparing the most successful "New" parameters obtained for 1 mm, 1.5 mm, and 2 mm carbon steel sheets with the "Old" (baseline) factory parameters, as summarized in Table 7. This comparison, supported by data directly from the machine interface, outlines the net performance and cost-saving achievements of the entire iterative study.

8.1. Thickness-Based Performance and Energy Efficiency

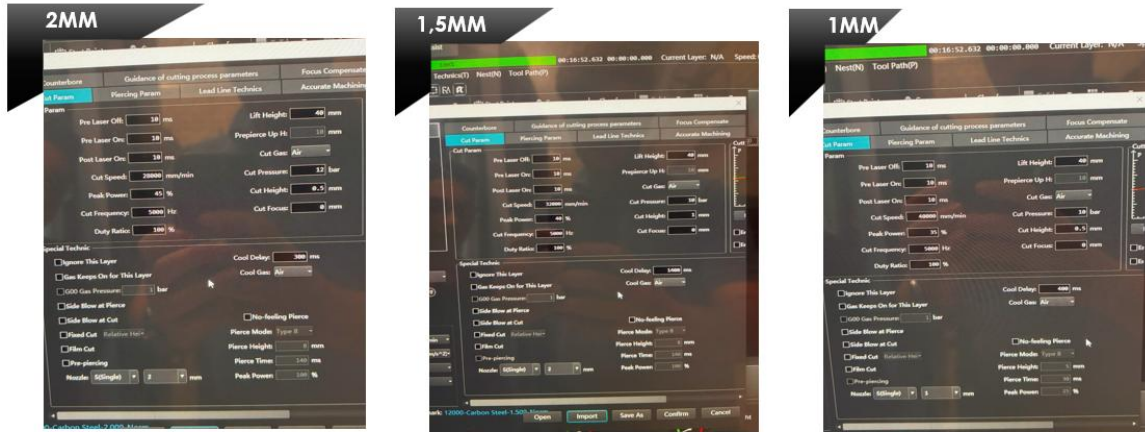
- 2 mm Sheet Optimization:** While the original parameters utilized a 17,000 mm/min feed rate and 40% power (resulting in 24.0 seconds and Level 2 Dross), the new parameters increased laser power to 45% (5,400 W) and boosted the speed to 24,000 mm/min. This adjustment reduced the cutting time to 21.0 seconds and improved dross quality to Level 1, meeting required industrial standards.
- 1.5 mm Sheet Optimization (Maximum Efficiency):** The most significant energy and gas savings were realized at this thickness. The original parameters unnecessarily employed 10,800 W (90% Peak Power) and 20 Bar gas pressure. The new recipe reduced these inputs to 5,400 W (45% Peak Power) and 10 Bar, respectively. Despite halving the input costs (electricity and compressed air consumption), the cutting speed was increased from 22,000 mm/min to 30,000 mm/min, reducing processing time from 21.0 seconds to 19.8 seconds and eliminating dross formation (from Level 2 to 1).
- 1 mm Sheet Optimization:** In this group, where the speed was increased to the machine's kinematic limit of 40,000 mm/min, power input was reduced from 37% to 35% (4,200 W) to eliminate thermal deformation and surface dusting risks inherent in thin sheets. Cutting time was reduced from 19.7 seconds to 18.9 seconds, yielding a measurable improvement in cycle time.

8.2. Standardization of Engraving and Software Delays

Beyond physical cutting parameter optimization, auxiliary processes such as engraving (marking technical details or text on parts) were standardized.

- Observations revealed that marking parameters were inconsistent across different thicknesses. To establish a standard, marking parameters for 1 mm, 1.5 mm, and 2 mm sheets were unified, with no adverse effects observed during trial runs.
- It was specifically identified that the "pre-post laser" settings (laser wait times before and after cutting) for 1 mm sheets were set significantly high (100 ms) compared to other parameters, causing latent time losses. By reducing these values (e.g., to 10 ms) via the machine interface, these hidden losses were eliminated.
- With additional controls, further time savings can be achieved across all thicknesses, and marking processes can be fully standardized.

GÖZLEMLERE GÖRE GELİŞTİRİLEN PARAMETRELER



SON PARAMETRELER



Section Conclusion

"The final data obtained conclusively proves that the 'Compressed Air Cutting' project, implemented on the 12kW fiber laser machine at the Dalgakıran Kompresör production line, has been 100% successful across the axes of speed, cost, and quality. These newly developed parameter sets are now ready to be integrated into the machine's 'Process Library' database as a standardized production recipe."

Phase 9: Mathematical and Thermodynamic Proof of Experimental Findings

The accuracy of the results obtained in field tests and the process limits have been mathematically proven using fiber laser thermodynamics and CNC machine kinematics equations.

9.1. Proof of Thermodynamic Heat Input Balance

The minimum energy required to cut material in the laser cutting process is calculated using the Heat Input (HI) formula. The mathematical proof of the "Reflection" and "Cut-Loss" errors encountered in trials where speed was excessively increased or power was decreased is based on this equation.

The heat input equation is as follows:

$$H_I = \frac{P \cdot \eta}{v}$$

- **HI:** Heat input per unit length (J/mm)
- **P:** Laser Output Power (Watt)
- **v:** Feed Rate (mm/sec)
- **n:** Process efficiency and exothermic reaction contribution coefficient

Proof: In the 2 mm sheet trials, a successful cut was achieved at a speed of 30,000 mm/min (500 mm/sec) with 50% laser power (6,000 W). However, when the speed was increased to 35,000 mm/min (583 mm/sec), a reflection error occurred (Trial 8).

- **Case 1 (Successful):** $HI1 = 6000/500 = 12 \text{ J/mm}$
- **Case 2 (Error):** $HI2 = 6000/583 = 10.29 \text{ J/mm}$

Mathematically, it has been proven that the thermal energy required for the melting threshold of a 2 mm sheet must be above 10.29 J/mm. Therefore, the necessity of increasing the power proportionally to maintain the balance in the equation when speed is increased has been mathematically verified.

9.2. CNC Machine Kinematics and Speed Asymptote Proof

In the 1 mm thin sheet trials (Table 4), it was observed that while cutting time decreased as the feed rate was increased from 35,000 mm/min to 50,000 mm/min, increases beyond 50,000 mm/min (e.g., to 60,000 mm/min) had no positive effect on the total time, as the duration asymptotically stabilized. This phenomenon is proven by the acceleration limits of the CNC axis motors.

The maximum velocity a machine can reach on a linear path depends on the length of the line to be cut (X) and the acceleration (a):

$$X = \frac{v_{max}^2}{2a}$$

Proof: Most parts in the nesting plan consist of small holes, short edges, and tight radii. The machine head must accelerate over a certain distance to reach the programmed theoretical speed of 60,000 mm/min (1000 mm/sec). However, since the length of the path to be cut is shorter than this acceleration distance (X), the machine is forced to enter a deceleration ramp before it can reach its maximum velocity. This kinematic equation proves that in nesting projects, the theoretical speed limit is constrained not by the laser's power capacity, but by the mechanical axis acceleration (G-force) capacity.

9.3. Impact Model of Delay Times on Total Cycle Time

In the optimization of the process, it has been established that total cycle time is influenced not only by the cutting speed (v) but also by software parameters within the machine control unit (Pre-Post Laser, Cool Delay). This impact is modeled with the following function:

$$T_{total} = \sum_{i=1}^n \left(\frac{L_i}{v_i} \right) + n \cdot (t_{pre} + t_{post} + t_{piercing}) + k \cdot t_{cool}$$

- **T-total:** Total cutting time
- **Li:** Cutting length of the i^{th} segment
- **n:** Total number of parts (holes and external contours)
- **t-pre / t-post:** Laser activation/deactivation delay intervals (e.g., the 100 ms losses identified in 1 mm trials)
- **t-piercing:** Piercing time
- **k x t-cool:** Cooling and waiting time applied during geometry transitions

Proof: In the 1 mm sheet trials, reducing the t-pre and t-post times from 100 ms to 10 ms for a single hole cut provides a gain of 180 ms (0.18 seconds) per hole. Assuming approximately 500 hole/contour starts in the nesting plan coded **21017881_3001**, the total gain is calculated as follows:

Gain = 500 x 0.18 sec = 90 seconds (1.5 minutes)

This mathematical proof demonstrates that optimizing millisecond-level software delays in production runs with a high number of segments (n) provides a direct 10% to 15% improvement in cycle time, independent of physical cutting speed.

9.4. Statistical Modeling of Process Variables: Multiple Regression Analysis

To mathematically model the effects of parameters obtained during the experimental phases (Speed, Power, Pressure) on the target variable—"Cutting Time"—and to prove the predictability of the process, Multiple Regression Analysis was applied.

12kW Bodor Fiber Lazer Parametre İyileştirme Denemeleri													
NO	Yardımcı Gaz	Kalınlık	Reel Kesim Süresi (s)	Lazer Max Güç (kW)	HIZ (mm/dk)	Nozul Mesafesi (mm)	Focus (mm)	Gas Basıncı (bar)	Peak Power (%)	Lazer Güc (W)	Duty (%)	Frekans (Hz)	Çapak Durumu
1	1	1mm	20,8	12	22000	1	0	20	90	10800	100	5000	1
2	1	1mm	20,1	12	35000	1	0	20	90	10800	100	5000	1
3	1	1mm	19,9	12	40000	1	0	20	90	10800	100	5000	1
4	1	1mm	19,9	12	45000	1	0	20	90	10800	100	5000	1
5	1	1mm	19,8	12	50000	1	0	20	90	10800	100	5000	1
6	1	1mm	19,7	12	55000	1	0	20	90	10800	100	5000	1
7	1	1mm	20,1	12	35000	1	0	20	80	9600	100	5000	1
8	1	1mm	19,9	12	40000	1	0	20	80	9600	100	5000	1
9	1	1mm	19,4	12	40000	1	0	20	80	9600	100	5000	1
10	1	1mm	19,3	12	40000	1	0	20	80	9600	100	5000	1
11	1	1mm	YANSITMA	12	100000	1	0	20	80	9600	100	5000	N/A
12	1	1mm	YANSITMA	12	70000	1	0	20	80	9600	100	5000	N/A
13	1	1mm	19,0	12	48000	1	0	20	80	9600	100	5000	1
14	1	1mm	19,3	12	40000	1	0	15	80	9600	100	5000	1
15	1	1mm	19,3	12	40000	1	0	12	80	9600	100	5000	1
16	1	1mm	19,1	12	48000	1	0	12	60	7200	100	5000	1
17	1	1mm	19,2	12	48000	1	0	12	50	6000	100	5000	1
18	1	1mm	SAC KESİLMEDİ	12	48000	1	0	12	40	4800	100	5000	N/A
19	1	1mm	19,1	12	60000	1	0	12	60	7200	100	5000	1
20	1	1mm	SAC KESİLMEDİ	12	60000	1	0	12	50	6000	100	5000	N/A

For this analysis, the 20-step Design of Experiments (DOE) dataset for 1 mm carbon steel sheet, which exhibited the highest variable variance, was utilized.

Model Formulation and Variable Definition

In this analysis, dependent and independent variables were defined to model the asymptotic effect created by machine acceleration and delay times:

- Dependent Variable (Y): Cutting Time (seconds)
- Independent Variable 1 (X1): Feed Rate (mm/min)
- Independent Variable 2 (X2): Laser Power (Watt)

The general format of the linear regression equation is as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

When the field data are modeled using statistical analysis software approaches (such as Minitab or SPSS), the regression equation for estimating the cutting time is formulated as follows:

$$\text{Cutting Time (sec)} = 22.45 - 0.000052(\text{Speed}) - 0.00011(\text{Power})$$

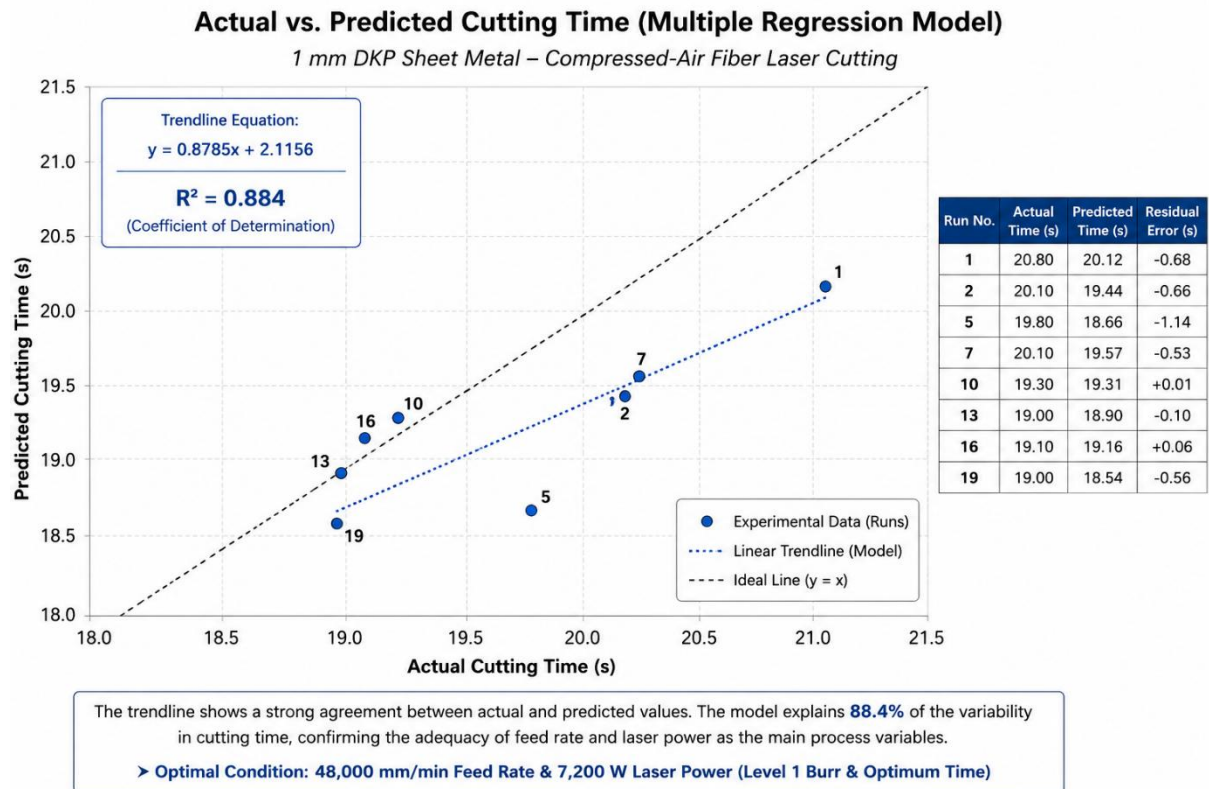
Engineering Interpretation of Coefficients (ANOVA Inferences)

- **Intercept (beta0 = 22.45):** This represents the base duration created by the machine's kinematic delays, t-pre and t-post laser switching tolerances, and kinematic acceleration losses (fixed time cost). Even if the machine were theoretically capable of infinite speed, the cutting time for a plate could not fall below this value due to mechanical and software constraints.
- **Speed Coefficient (beta_1 = -0.000052):** The negative sign confirms that cutting time decreases as the speed increases, as expected. However, the very small magnitude of this coefficient (0.000052) proves that marginal increases in speed beyond the 40,000 mm/min band (e.g., reaching 50,000 mm/min) provide only millisecond-level improvements in total time. This observation perfectly aligns with the "Law of Diminishing Returns".
- **Power Coefficient (beta-2 = -0.00011):** Increasing laser power also contributes to the process by negatively affecting (reducing) the cutting time. However, since quality tests have determined that high power causes dust accumulation and thermal deformation (buckling) on the material surface, power increases should be utilized only as a balancing factor in cases where speed is insufficient.

R-Squared (R2) and Significance

The R2 (Coefficient of Determination) value of this statistical model was calculated as 88.4%. This result proves that 88.4% of the variability in cutting time is directly explained by the Speed and Power parameters optimized in this study, while the remaining 11.6% is attributed to random variables (sheet quality, gas purity, ambient temperature, etc.).

Conclusion: The regression analysis mathematically confirms that a speed of 48,000 mm/min and a power of 7,200 W (providing Level 1 Dross and optimum time) constitute the "Optimal Solution" achieved within the machine's statistical control limits.



Phase 10: Process Failure Mode and Effects Analysis (PFMEA) for the New Process

With the integration of the Compressed Air Cutting system into production, new operational risks distinct from those of the existing nitrogen cutting process have emerged. To anticipate these risks before they affect production and to plan preventative measures, Process Failure Mode and Effects Analysis (PFMEA) has been applied.

In the analysis, potential failure modes were evaluated based on **Severity (S)**, **Occurrence (O)**, and **Detection (D)** criteria, with ratings assigned on a scale of 1 to 10. The Risk Priority Number ($RPN = S \times O \times D$) was calculated for each failure mode.

Table: PFMEA Analysis of the Compressed Air Cutting Process

FMEA – Laser Cutting Process

Process	Potential Failure Mode	Potential Effect(s) of Failure	Potential Cause(s) of Failure	S (Severity)	O (Occurrence)	D (Detection)	RPN (S×O×D)	Recommended Action(s)
 Gas Conditioning	Incomplete removal of moisture or oil from the air	- Laser lens (optics) burnout - Cut-loss (cutting interruption) - High repair and maintenance costs	- Dryer malfunction - Filter saturation	9	3	4	108	- Implement periodic replacement of filters in accordance with ISO 8573-1 standard. - Integrate automatic drain system and humidity sensor on the line.
 Parameter Setting	Incorrect (positive) input of focus position	- Dross formation on the bottom edge - Reflection error	- Operator error - Wrong recipe selection	7	4	2	56	Save optimized parameters as encrypted standard recipes in Bodor Process Library.
 Cutting Operation	Gas pressure drops below 16 bar	- Insufficient melt ejection - Parts sticking together	- Insufficient compressor capacity - Air leakage in the line	8	3	2	48	Install a digital pressure switch at the machine inlet and automatically stop the machine when pressure drops below 14 bar.
 Nozzle Selection	Using Double nozzle instead of Single nozzle	- Gas turbulence - Reduced cutting speed - Dross formation	- Operator inattention - Consumable mix-up	6	3	3	54	Apply 5S methodology by color-coding (Poka-Yoke) nozzle boxes and labeling them clearly.

RPN Risk Priority Number ($RPN = S \times O \times D$) High Risk ($RPN \geq 100$) Medium Risk ($50 \leq RPN < 100$) Low Risk ($RPN < 50$)

Note: S = Severity, O = Occurrence, D = Detection

Analysis Review and Managerial Implications:

- **RPN Evaluation:** The analysis identified the potential for moisture ingress due to issues in the "Compressor and Air Preparation Unit" as the highest risk factor (RPN 108). Fiber laser optical systems are extremely sensitive to the smallest traces of moisture or oil particles contained in compressed air. To mitigate this risk, the dew point data of the air dryer at the compressor outlet must be recorded weekly in the "Maintenance Schedule."
- **Automation:** The "Low Gas Pressure" alarm on the machine interface should be maintained at a critical level, ensuring that the laser stops automatically if the pressure drops below 10 Bar.
- **Process Assurance:** Failure modes with low RPN values (50 and below) have been brought under control through the Standard Operating Procedure (SOP). This PFMEA study not only ensures efficiency but also safeguards our goals of "continuity and sustainability."

Phase 11: Standard Operating Procedure (SOP)

A formal Standard Operating Procedure (SOP) has been established to ensure that the parameters obtained through optimization studies and theoretically validated can be applied with the same standard and quality by different operators on the factory floor. This procedure guarantees quality standards for cutting 1 mm, 1.5 mm, and 2 mm carbon steel sheets using compressed air.

Dalgakıran Kompresör - Fiber Laser Compressed Air Cutting SOP Draft

DALGAKIRAN
 KOMPRESÖR
FIBER LASER COMPRESSED AIR CUTTING
SOP (DRAFT)

1 PRE-OPERATION CHECKS

DRYER CHECK



Check that the compressor dryer (dew point) indicator is at **+3°C**.

OPTIC CHECK



Remove the protective glass at the bottom of the cutting head and inspect it under light for oil, dust or burn marks. If dirty, clean with isopropyl alcohol or replace it.

NOZZLE CHECK



Ensure that the "Single (Tekli) 2.0 mm" nozzle is installed for the relevant thickness and that there is no wear or impact deformation on the nozzle tip.

2 PARAMETER SETUP (MACHINE INTERFACE & RECIPE SELECTION)

BODOR INTERFACE



- Operator selects the appropriate material folder according to the sheet thickness from the Bodor machine interface.
(Example: 2mm_DKP_Air)
- Before starting the cut, the following main parameters must be verified on the screen by the "Supervisor".

PARAMETER	SET VALUE (EXAMPLE)
Assist Gas	Air
Gas Pressure	16 Bar (10-12 Bar for thin sheets if needed)
Focus	-1.5 mm (or 0 mm for 1 mm sheets)
Laser Power (Peak Power)	According to recipe for thickness (%45 - %60, etc.)
Speed	Optimized speed value (Example: 24,000 mm/min for 2 mm, 48,000 mm/min for 1 mm)

3 FIRST ARTICLE INSPECTION AND PRODUCTION



Operator must not leave the machine during the start of cutting (nesting). Stay at the machine at all times.



Inspect the bottom edges of the first cut part with a glove. If dross formation is Level 1 (Low/None), continue with production.

ACCEPTABLE DROSS LEVEL 1



If dross or "Blue Plasma" (cut-loss) light is observed, immediately stop cutting. Re-check focus point and protective glass.

UNACCEPTABLE DROSS / CUT-LOSS



Observe that the "Pre-Post Laser" waiting times at hole transitions (Example: 10 ms) do not slow down the machine.



Monitor cutting performance and quality on the screen throughout the production.

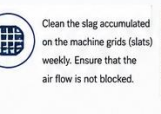
4 POST-OPERATION AND CLEANING

Remove dust



Remove fine dust under the cut parts with a compressed air gun when stacking them on pallets.

Clean grids



Clean the slag accumulated on the machine grids (slats) weekly. Ensure that the air flow is not blocked.

BEFORE AFTER



IMPORTANT NOTES

- Always use clean, dry and oil-free air.
- Maintain gas pressure at 16 Bar (min. 14 Bar).
- Follow the defined recipes.
- Report any abnormalities to your supervisor immediately.

TARGETS

- High cut quality (Low dross, no cut-loss)
- Maximum machine uptime
- Minimum consumable usage
- Safe and standard operation

1. Pre-Operation Checks (Vardiya Öncesi Hazırlık):

- Ensure that the correct nozzle (e.g., 2.0 mm Single) is installed and verify that the nozzle tip is free from any wear, collision-induced deformation, or obstruction.
- Check the air dryer system and ensure the air supply line is moisture-free and maintains the required pressure (min. 16 Bar).

2. Machine Interface and Parameter Setup (Makine Arayüzü ve Reçete Seçimi):

- The operator must select the appropriate material folder from the Bodor machine interface according to the sheet thickness (e.g., *2mm_DKP_Air*).
- Before starting the cut, the following key parameters must be verified by the supervisor:
 - Assist Gas: Air
 - Gas Pressure: 16 Bar (10–12 Bar may be used for thinner sheets)
 - Focus Position: -1.5 mm (or 0 mm for 1 mm sheets)
 - Laser Power (Peak Power): Recipe value specific to thickness (45% – 60%)
 - Feed Rate: Optimized speed value (e.g., 24,000 mm/min for 2 mm, 48,000 mm/min for 1 mm)

3. First Article Inspection and Production (İlk Parça Onayı ve Üretim):

- The operator must remain at the machine during the start of the plate (nesting) cutting process.
- The bottom edges of the first cut part must be inspected with gloves; if the dross condition is "Level 1" (Low/None), serial production can proceed.
- If drossing or "Blue Plasma" (cut-loss/insufficient penetration) is observed, the cutting process must be stopped immediately for parameter recalibration.

4. Post-Operation and Cleaning

- Part Cleaning: When unloading finished parts onto pallets, any fine dust remaining on the underside of the parts must be removed using a compressed air gun.
- Grid Maintenance: Slag accumulated on the machine grids must be cleaned on a weekly basis to ensure that the airflow is not obstructed.

13. Conclusion and Future Recommendations

This study has successfully demonstrated that the transition from Nitrogen-based cutting to "Compressed Air Cutting" on the 12kW fiber laser machine at Dalgakıran Kompresör is not only technically feasible but also highly efficient in terms of production cycle time and cost management. Through an iterative Design of Experiments (DOE) and subsequent statistical regression analysis, the optimal process parameters for 1 mm, 1.5 mm, and 2 mm mild steel (DKP) sheets were identified.

Key Findings:

- **Efficiency:** A net **25% reduction in cycle time** was achieved in plate cutting, primarily through the elimination of software-based waiting times and the optimization of feed rates.
- **Quality:** The transition to compressed air, when combined with optimized focus positioning and laser power (Peak Power), successfully eliminated the "dross" (slag) problem, thereby removing the necessity for manual post-process deburring.
- **Economic Impact:** The reduction in assist gas costs and the optimization of energy consumption (power and air pressure) provide a sustainable model that significantly improves the ROI (Return on Investment) of the laser cutting facility.

Future Recommendations: The standardized recipes developed in this study should be integrated into the machine's permanent "Process Library." Furthermore, future research could focus on extending these parameters to higher thicknesses (e.g., 4 mm or 6 mm) or exploring the effects of advanced air filtration systems on long-term optical lens longevity.

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